Navigating the German EV Market: Strategies for BYD's Sales Growth

Master Thesis, MSc in Business, Language and Culture - Diversity and Change Management

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Abstract

With the escalating global attention on climate change and environmental concerns, electric vehicles (EVs) have emerged as a crucial solution for mitigating CO2 emissions, driving a surge demand worldwide. As the leading EV manufacturer and having solidified its leadership in the Chinese market, BYD has accelerated its expansion efforts overseas. However, its entry into the German market with electric passenger vehicles at the end of 2022 has yielded disappointing sales over the past year. Thus, this master thesis aims to explore *How can BYD adjust its business strategy adapting to the German EV market to increase sales?* The analysis addresses three sub-questions: the institutional opportunity and challenge does BYD face through the lens of institution-based view, the opportunities and challenges BYD face in the German EV market is investigated by applying Porter's Five Force and VRIO frameworks is used to identify the resources can help BYD achieve a competitive advantage in the German EV market. Throughout the analysis, key considerations for BYD's strategic adjustment in Germany include enhancing infrastructure with government collaboration to address shortcomings in current EV policies, boosting brand awareness and loyalty, maintaining its R&D advantage to develop cutting-edge technologies while ensuring cost control, and adopting market segmentation strategies to tailor objectives and marketing approaches for different segments. These adjustments are crucial for BYD to expand its customer base and increase sales in the German EV market.

Keywords: Electric vehicle, BYD, Business strategy ,Electric vehicle, Institution-Based View, Porter's Five Forces, VRIO framework.

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List of Abbreviations

AFIR=Alternative Fuels Infrastructure Regulation

AI= Artificial intelligence

BEV= Battery electric vehicle

DJI= Da-Jiang Innovations

EV= Electric vehicles

EVs= Electric vehicles

FCEV= Fuel cell electric vehicles

IBV= Institution-based view

NCM= Lithium nickel manganese cobalt oxides

NCM= Lithium nickel manganese cobalt oxides

NPF=National Policy Framework

PHEV= Plug-in hybrid electric vehicles

RBV= Resource-based view

WLTP= Worldwide Light Vehicle Test Procedure

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Chapter 1 Introduction

1.1 Introduction

Along with increasing awareness of climate change and the environment around the world, and attention being paid to the challenges of environmental pollution and energy depletion, a new era of technological innovation and industrial transformation is unfolding worldwide.

The convergence of automobiles with energy, transportation, information, and communications technologies is rapidly progressing. Electrification, connectivity, and intelligence are becoming dominant forces in the automotive sector. Among the key developments in this industry are the rise of electric vehicles (EVs), which capitalize on alternative energy, advanced materials, and technologies like the Internet, big data, and artificial intelligence. These advancements are driving the optimization of energy usage and the evolution of smarter transportation systems. After signing of "Paris Agreement", countries have formulated corresponding strategic plans and policy mechanisms in promoting the development of EVs. This has led many large multinational automobile companies to actively invest EVs, strengthen R&D (research and development) of EV (Electric vehicle) investment and enhance capabilities in the EV industry. (McKinsey, 2021)

China has made rapid progress in electric vehicles. Through continuous efforts, China's production, sales and ownership have maintained a leading position in the world for five consecutive years since 2015. By 2017, the cumulative number of vehicles in stock exceeded 2 million., sales have reached 2.7 million vehicles in 2021, becoming a major country in EV, with significant technological progress in the area such as as batteries, motors, and electronic control technologie and enhanced industrial competitiveness. Additionally, substantial progress has been made in infrastructure development forming the world's largest charging and swapping network, by 2021china has formed the world's largest charging and swapping network, with a total of 5.21 million charging piles and 1,973 power swap stations established.

These provide prerequisites for Chinese EV companies to go overseas. (Salaberria & Xia, 2023)

Germany is the birthplace of the world's automobile industry and the largest automobile market in Europe. It has well-known manufacturers such as Daimler, Volkswagen, and BMW. Germany also has many auto parts giants such as Schaeffler Group, Mahle, Deutz, Getrag which are among the top players in the field of auto parts. There are 46 million passenger cars on the road every year. With the development of the situation and the transformation of automobile maintenance, the German automobile industry is also promoting the production and use of EVs. In 2019, Germany achieved a significant milestone that surpassed Norway for the first time and became the largest EV registration market in Europe. The country sold 700000 electric cars in 2023, surpassing all other European markets. In terms of infrastructure, Germany invested 3.5 billion euros to bolster the construction of public electric vehicle charging stations. (COLLINS, 2024) Currently there are more than one hundred thousand public electric charging stations in Germany and their plan is to achieve 1 million public EV charge points by 2030. (Dirkx, 2023)

BYD, a Chinese domestic car brand, is the leader in the field of EV. In 2011, the BYD E6 was born, the world's first mass-produced and sold pure electric vehicle. Eight months earlier than Tesla's Model S, the E6 became the world's first, laying the foundation for BYD's global expansion. In August 2022, by signing a contract with the European dealer Hedin Mobility Group, BYD marks its official entry into the German EV market and signifying an important move to expand its presence in Europe. the sales of BYD electric vehicles in Germany grew very slowly. In 2023, the total unit sales of BYD's fully electric vehicles were only 4139. (Opletal, 2024) which accounted for less than 1% of electric vehicle sales in Germany and It is far from BYD's sales target of capturing 5–10% market share and is way behind compared to Tesla and Volkswagen. It is also a huge difference comparing to BYD in China where the unit sold of BYD electric vehicles achieved over a million.

This situation rise interest for research to understand what the reason behind the BYD's poor performance in the German EV market and what should BYD do to improve it? Therefore, this research delves into case studies in the feild of EV, focusing on China's leading EV company, BYD. The paper aims to answer the research question: **How can BYD adjust its business strategy adapting to the German EV market to increase sales?**

In order to provide the comprehensive understanding and answering the research question is subject to addressing the following sub-questions:

I. What institutional opportunity and challenge does BYD face in Germany?
II. What opportunities and challenges does BYD face in the German EV market?
III. Which resources can BYD exploit to achieve a competitive advantage in the German EV market?

In order to answering the sub-questions, the Institution Based View is applied to explore the institutional opportunity and challenge does BYD face in Germany. Porter's Five force is addressed to gain the insight of the opportunities and challenges BYD face in the German EV market, Finally, Resource Based View will base on the VRIO framework to investigate which internal resources of BYD could be an competitive advantage to help company increase sales in the German EV market. By answering the research question this thesis also aims to offer insightful implications to other Chinese EV companies who seeking to enter the German and this study will attach to the academic field of managing dynamic organizations in a global environment with a focus on institutions, culture, and strategic complexity.

1.2 Delimitation

Three major delimitations are identified in this thesis,

1) The scope of this thesis is limited to BYD's passenger electric vehicle (BEV and PHEV)
2) The focus is on German EV market.

The reason for the first delimitation is that 70% of BYD's total revenue stemming from the automobile business and BYD places significant emphasis on the development of its fully passenger car. The electric vehicles are also the key to the carbon-reduction of the automotive industry and are in line with future development trends. BYD's electric vehicles experiencing slowly market progress in Germany currently. Given these factors, investigating the BYD's passenger electric vehicle business holds substantial relevance for its ongoing development. (BYD, 2024)

The reason for the second delimitation is that Germany is the birthplace of the world's automobile industry and the largest automobile market in Europe and also the largest car market in Europe, there are 46 million buses on the road every year. The German EV market continues to grow, the sales of EV cars have climbed each year since 2016, the country sold 700000 units EV in total, which include 524000 units of BEV and 176000 units of PHEV in 2023, (COLLINS, 2024) surpassing all other European markets. Secondly, expanding into the European market is an important part of BYD's overseas expansion strategy, and Germany is one of the first countries for BYD to enter the European market. Therefore, studying the German EV market holds great significance for BYD's strategic expansion in Europe.

1.3 Key definition

1.3.1Competitive advantage

Porter and Barney both have contributed to the concept of "competitive advantage" significantly. According to Porter "competitive advantage" is the capacity acquired through distinctive attributes and resources to outperform others within the same industry or market. (Porter, 2008)

Barney has described "competitive advantage" as a company deemed to possess a competitive advantage when it is executing a strategy that creates value and is not being pursued by any current or potential competitors at the same time. The competitive advantage may be temporary and can be eroded over time as competitors imitate or surpass the company's strategies. (Barney, 1991)

1.3.2Sustained competitive advantage

Sustained competitive advantage represents a lasting form of superiority in the market, enduring over an extended timeframe. It entails upholding a leading position through elements that are challenging for competitors to duplicate or counteract, such as proprietary technology, robust brand allegiance, or exclusive access to crucial resources. This enduring advantage empowers a company to sustain its dominance and profitability over the long haul. (Barney, 1991)

In this thesis, the competitive advantage pertains to BYD's internal resources and competences. Through the VRIO examination, BYD’s internal resources and competences in which could be considered as the sustained competitive advantage is identified. This is pivot for BYD increase EV sales in Germany.

1.4 Content and Thesis Structure

Chapter I – Introduction

The objective of chapter I is to introduce the thesis by provide brief background information, describing the motivation, objective, and importance of this research. The research question and sub-questions are presented in this chapter, as well as delimitation and key definition. This will allow reader to understand the underlying grounds for choosing the research area and topic.

Chapter II – Literature review

Chapter II contain the literature reviews that focus on discuss the relevant literature, including the regulation and policies of EVs, the EV infrastructure in Germany, and the German EV market, The aim of the literature reviews in this chapter is to enhance the understanding of the research topic. Research gap and significance are represented at the end of this chapter.

Chapter III – Theoretical Framework

The critical concepts and theories are interpreted in this chapter, such as the institution-based view, the industry-based model of Porter's Five Forces, the resource-based view, and the VRIO framework. Those theoretical frameworks are a foundation for research question and used to understand the research problem and guide the development and analysis of this research.

Chapter IV - Methodology

In this chapter discusses the methodology of this study. research philosophy will be presented here, and the validity of data collection is also explained. Furthermore, the description section will also be included which contains more background knowledge of research objective and the research and the reason for the selection of BYD as a case company for this research The limitation of the research methodology is revealed in the Chapter of Discussion.

Chapter V – Analysis

The fifth chapter will provide an analysis that uses theories and models in the theoretical part and combines the relevant data collected together to conduct interpretation in-depth analysis of each sub-question. Analysis of each sub-research question with corresponding sub-conclusions. This chapter serves as the knowledge foundation, which could help readers to have a better understanding of the research findings in the next chapter.

Chapter VI – Finding and Discussion

This chapter contains two sections, the first section will discuss the findings throughout the analysis. The second section and the second part contain a discussion aim to provide the proper suggestion for BYD based on the analysis and findings. These two sections together should form ultimately answer to the research question.

Chapter VII Conclusion and Further Perspective

The final chapter will summarize the aspects considered in this thesis and the limitations of the study. In closing, the attention drawn to several ideas concerning future research relate to this topic.

Chapter 2 Literature Review

This chapter discuss the existing literature relevant to the research scope which include the review of relevant study on regulation and policies of EVs, the EV infrastructure in Germany, and the German EV market. The literature reviews shed light on identifying major research areas and pointing out the gaps in existing research, and thus lead the way to this study.

2.1 Regulation and policies of EVs in Germany

As we all know, the development of electric vehicles (EV) It is greatly affected by external factors. Danny Kreyenberg's paper "Analysis of Electric Vehicle Market Penetration in the German Market to 2030" describes how various conditions, including EV costs, government incentives and energy prices, affect the sales trajectory of electric passenger cars in Germany.

This paper mentioned the German government's determination to develop electric vehicles. Since the establishment of the electric vehicle platform in 2010, Germany has incorporated the development of electric vehicles into national policies and invested huge energy and financial resources in response to the EU's carbon emission reduction policy. Emissions and indicators to address the energy challenge. Kreyenberg's research, this article takes an in-depth look at the current status and expected progress of various types of electric vehicles in Germany.

In the literature, author used VECTOR21 vehicle technology scenario model is used to model the fleet in three scenarios. In this analysis the parameters of external conditions such as governmental support/subsidies for electric vehicles and energy prices as well as internal (OEM) influencing factors such as the mark-ups for sold vehicles, In the reference scenario with first business-as-usual scenario, given the absence of monetary incentives for electric vehicles in combination with an (only) moderate increase in the oil price, the anticipation shows that, the current market share of car industry in Germany keep also constant, so even though non plug-in hybrids gradually replace conventional combustion engine drivetrains, nearly 50% consumer will still choose diesel or gasoline. In the reference scenario with business-as-usual parameters. (Propfe, Kreyenberg , Wind, & Schmid, 2013)

In the second scenario, the author has adjusted given parameters with purchase price incentives from 5000 EUR, high oil prices, and low prices for hydrogen, the electricity will still not able to reach the target of 1 million EV on road in 2020 and 6 million in 2030, In the

last scenario with substantial OEM mark-up reductions and external conditions as in the business-as-usual scenario, 3.28 million on road, however, this seem to be less realistic on account of the long payback periods. (Propfe, Kreyenberg , Wind, & Schmid, 2013)

The conclusion from the literature shows us how important external factor would impact sales of EV car in Germany that analysis shows that the market success of alternative powertrain vehicles depends to a great extent on external conditions. Without purchase price incentives, rising oil prices, and low energy costs for hydrogen and electricity, alternative powertrains will not enter the market. One problem might be that the cost of fuel cells and batteries will

remain on a high level unless they can be made in mass production on a large scale. We therefore use the experience curve approach to model the cost of the different powertrain technologies in the future The previous article showed us the importance of external conditions for electric vehicles. One of the reasons BYD has been able to grow and develop is the support of the Chinese government. Public transportation in many cities has signed purchase contracts with BYD public trams. Therefore, for byd, the external environment of the German market also has a great impact on the development of byd in the German market. (Propfe, Kreyenberg , Wind, & Schmid, 2013)

In order to support the development of the German tram industry, Germany has invested heavily as mentioned above, and there are also corresponding incentive policies and legislation in Germany to promote the development of electric vehicles. Information released by the European Alternative Fuels Observatory understands that the current incentives and legislation in Germany cover various benefits, including ownership tax exemptions, which means that as long as they are battery electric vehicles (BEV) and fuel cells registered before December 31, 2025 Electric vehicles (FCEV) are exempt from ownership tax. In addition, vehicles emitting 95 grams of carbon dioxide per kilometer or less are exempt from the annual circulation tax. (Commision, 2024)

There are also company tax benefits, such as reductions in taxable benefits for the private use of company cars, and special incentives for lower-priced BEVs. Information released by the European Alternative Fuels Observatory also mentions various subsidies for the purchase of new and second-hand pure electric vehicles, with subsidies based on the purchase price and lease term. Other financial benefits include exemptions for private owners of plug-in electric vehicles to charge at employer premises, and tax adjustments provided by the German government for electric company cars. Infrastructure incentives include funding for the purchase of electric vehicles and charging infrastructure, with a particular focus on municipal and commercial fleets. Funding is also available for private charging stations in residential buildings. Local incentives may vary, and while the environmental bonus may be combined with other public funding programs, most are targeted at companies, municipalities, or associations rather than private individuals. (Commision, 2024)

2.2 EV Infrastructure in Germany

In addition, the development of electric vehicles requires state investment in corresponding infrastructure, such as charging stations. According to information released by the European Alternative Fuels Observer, the EU has set NPF (National Policy Frameworks)targets for the deployment of alternative fuel infrastructure in each member state. The NPF Directive requires member states to adopt a national policy framework and report to the European Commission by November 18, 2016, which includes national targets for each member state to deploy alternative fuels infrastructure.

Under the AFIR (Alternative Fuels Infrastructure Regulation)member states must ensure that the deployment of public charging stations matches the use of electric vehicles and provides sufficient power support. Specifically, public charging stations are required to provide minimum power outputs for different types of electric vehicles, including a total power output of at least 1.3 kW per electric light vehicle and at least 0.8 kW per plug-in hybrid light vehicle. output. (Transport, 2021)

European Union points out that the NPF also stipulates specific targets such as the rollout of electric vehicle charging infrastructure in urban and suburban areas and on the TEN-T core network(Trans-European Transport Network). In the NPF, Germany set infrastructure targets for 2025 and 2030, the goal in 2030 is 1 million electric refueling points. (Transport, 2021) And according to the Observatory's implementation tracking of for each of the fuel types, it shows that Q1 2024, Germany has a total of 125,997 recharging points, including 24,293 DC(direct current) recharging points and 171,704 AC(alternating current) charging stations. An year on year growth rate is almost 32%, and the charging capacity has increased by more than 40%. Among them, there are 93,343 unrestricted access charging stations and 32,625 restricted access charging stations, such as charging points in large warehouses or convenience store parking lots, underground parking lots, hotel and restaurant venues, etc. The opening hours are limited by business hours. (Transport, 2021)

AFIR fleet-based target attainment current status total power (kw) output reached 5000k, while the target kw is 2500k, The NIR target of BEV + PHEV fleet (M1 & N1 vehicle category) is 8,5 million by 2030, and now The completion status is only 2.5 million, the 2030 NIR target recharging infrastructure is 1 million, and the current recharging points are 125997. These results show that if Germany is to achieve these goals set by the EU, Germany still needs to increase investment in infrastructure and improve efficiency, otherwise It will be difficult to achieve these targets by 2030. (Transport, 2021)

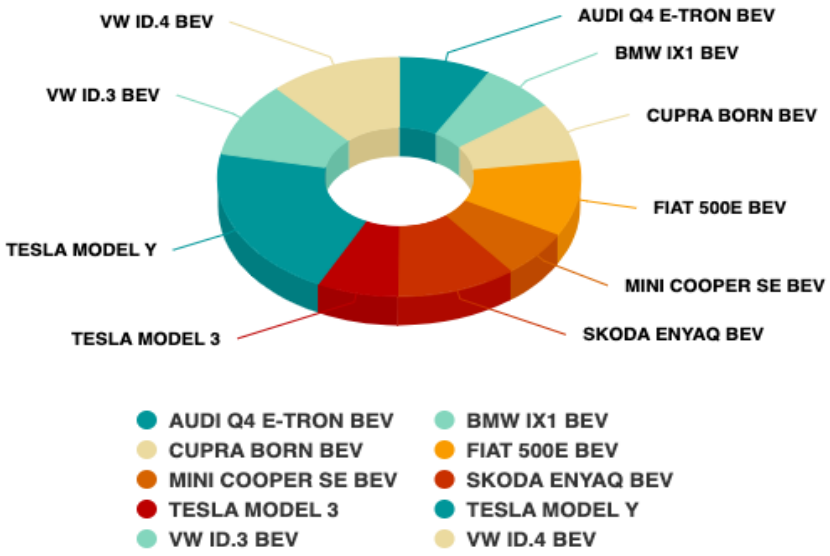
2.3 The German EV Industry

In response to the global crisis of climate change, nations worldwide are obligated to reduce carbon emissions. According to data from Statista, approximately one-quarter of the world's carbon dioxide emissions originate from the transportation sector, highlighting the urgent need to advance the electric vehicle (EV) industry. Electric vehicles, powered by renewable

energy sources, emerge as a dependable solution advocated by numerous countries to curtail carbon dioxide emissions and mitigate environmental pollution caused by traditional internal combustion engines. Market analysis provided by Statista reveals insights into the electric vehicle market, encompassing information about countries and regions with established public electric vehicle charging infrastructure. In this context, "public" denotes accessibility to charging infrastructure without constraints. Vehicles equipped with batteries or classified as plug-in hybrid electric vehicles (PHEVs) are categorized as electric vehicles. All primary data presented represent the sales volume and fundamental configurations of new vehicles for the respective year, excluding sales of pre-owned cars and modifications made to newly sold vehicles. The pricing and revenue figures provided are based on the standard model. (Statista, 2024)

The electric vehicle market is segmented into two main categories: battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs). This classification accounts for the distinct features and market penetration rates of each electric vehicle type, offering a nuanced understanding of market dynamics. Prioritizing new vehicle sales and their standard configurations ensures transparency, while excluding pre-owned and customized vehicles maintains focus on the evolving landscape of the electric vehicle industry.BEV and PHEV related market as passenger car,motocycles,shared mobility,travel tourism,bicyles, third party logistics, commercial verticles, the data from European Commission that in the total registration volume of German light-duty vehicles (m1+n1) reached over 49 million in 2023. The total registration volume of German EV passenger cars were close to 3 million units that including BEV of 1.44 million units and PHEV of 1.03 million units, accounting for 5.97% of German light-duty vehicles market. (Commision, 2024)

The best-selling BEV models of 2023 in Germany are shown in Figure 1, including 45618 units of Tesla model Y, 15865 units of Tesla Model 3; 22270 units of VM ID3 and 22270 units of VM ID4. (Commision, 2024)



(Figure 1: the best-selling BEV models of 2023 in the German EV market,Source from the European Alternative Fuels Observatory)

Statista is optimistic about the development of the German electric vehicle market. Combining their views on the German environment and consumer preferences will promote the development of the German electric vehicle market. Highlight from the Statista point out the strong economic condition of country the subsidies from the government such as the tax benefits as mentioned earlier Coupled with the various government subsidy policies mentioned earlier for companies manufacturing electric vehicles, it will help reduce the selling price of electric vehicles ., Although Germany has been a country with a long-term demand for internal combustion engine vehicles, German

consumers still advocate environmental protection and green travel. Electric vehicles provide them with the opportunity to reduce environmental pollution and go green, and are enthusiastic about the development of the electric vehicle industry across the country, Europe and the world. The attention will also affect the concept of German consumers. So statista

believes that demand for electric vehicles is growing in both consumer and commercial sectors. (Statista, 2024)

Therefore, statista made the following predictions for the German electric vehicle market from 2024 to 2028 based on the above reasons and historical data collected from all parties. According to Statista, Germany is poised to experience substantial growth in the Electric Vehicles (EV) market. By 2024, the projected revenue is estimated to reach US$60.2 billion. Moreover, the market is expected to demonstrate a promising compound annual growth rate (CAGR) of 14.49% from 2024 to 2028. This growth trajectory is forecasted to drive the market volume to US$103.4 billion by 2028. It is anticipated that unit sales of electric vehicles in Germany will reach 1,721,000 vehicles by 2028. Regarding pricing, the volume-weighted average price of electric vehicles in the German market is expected to be US$60.1k in 2024. In the global market context, China is projected to lead in revenue generation, reaching US$319 billion in 2024. Germany, renowned for its robust automotive industry, is witnessing an uptick in demand for electric vehicles as consumers increasingly prioritize environmental consciousness. (Statista, 2024)

2.4 Challenges of German EV Market

However, many German scholars and media regarding the electric vehicle industry are currently not as optimistic as Statista. The journal" Prioritizing the Chicken or Egg? Electric Vehicle Purchase and Charging Infrastructure Subsidies in Germany" described that the effect of the German electric vehicle industry on reducing CO2 emissions is not obvious, because the continued growth of total private vehicle demand has kept the conversion rate between fossil fuel vehicles and electric vehicles stable, and the proportion of coal-fired power is still high at 38% in 2018. This once again confirms that the current penetration of the German electric vehicle market is insufficient. The key reasons involve capital investment and cost control. The first is that the development of the electric vehicle industry requires a large amount of capital to promote technological innovation and improve urban infrastructure.

These investments are often long-term investments that will not pay off in one or two years. Although European governments have developed incentive schemes for electric vehicles (European Alternative Fuels Observatory, 2020), as this involves changing the structure of charging infrastructure, fossil fuel stations are now spread across the country. With the transition to electric vehicles, in Germany, the grid not entirely decentralized, the scarcity of infrastructure in remote areas will also affect the speed of the electric vehicle transition. (Zink, Valdes, & Wuth, 2020)

The journal also points out that the transition to electric vehicles hinges on advancements in battery technology. Currently, Germany lacks a competitive edge in battery production, as Bosch is the only German company in engaged in this technology who already decided against pursuing investment projects in this sector in the spring of 2019. Battery technology is predominantly led by consumer product companies, with Samsung and Panasonic at the forefront. Additionally, the Chinese company CATL signed an investment agreement with the Thuringia state government in Berlin to establish Germany's largest lithium battery production plant. CATL's Thuringia battery production facility is primarily focused on research, development, and manufacturing of lithium-ion batteries. (Zink, Valdes, & Wuth, 2020) CATL Europe President Ma Matthias Zentgraf calls it "Western Europe's first large-scale battery production project." He introduced that one production line has been started in Arnstadt, and five more production lines will be put into operation in the future. It is expected to reach the initial production capacity target of 14GWh (gigawatt hours) in 2024. Its products will be used by world-renowned car companies such as BMW, Volkswagen, Daimler, Jaguar Land Rover, and PSA. It reduces the battery transportation cost and time of these car companies. (Zink, Valdes, & Wuth, 2020)

The Institute for Energy Research reveals the current situation of the German electric vehicle industry: Germany expects a 14% drop in electric vehicles in 2024. the obstacles to the growth of the electric vehicle market and the growth of electric vehicle sales. Slowdown and

uncertainty about future demand forecasts Government incentives for electric vehicles are weakening, and infrastructure and pricing remain significant obstacles to the full expansion of electric vehicles. (IER, 2024)

In terms of infrastructure, the article mentioned that there are currently less than 110,000 charging infrastructures in Germany. There is still a long way to go to achieve the previous strategic goal of 1 million charging stations by 2023 set by German Transport Minister Wiesing. To achieve the goal, Germany must double its current rate of infrastructure development.

In terms of pricing, the market of EV price above 35,000 euros is already saturated, and the categories of electric vehicles below 25,000 euros are very limited. Therefore, only high-income people in Germany can afford electric vehicles. In addition, Germany will cancel the purchase of electric vehicles in 2022. With a subsidy of 4,500 euros, The German Association of the Automotive Industry (VDA) expects that sales of electric vehicles will drop by 14% in 2024. If this situation continues for a long time, it will be difficult to achieve the German government's target of 15 million electric vehicles on the road in 2030. (IER, 2024)

Along with electric vehicles in Germany, automakers are adjusting their strategies to deal with the changing electric vehicle landscape. German car-hailing companies such as BMW and Volkswagen have slowed down their strategic goals to achieve full electrification. BMW CEO Oliver Zipse spoke at the Berlin Automobile Industry Association New Year reception. Zipse told the German government Taking a cautious approach to his previously proposed strategy of fully implementing electric vehicles by 2030, he has advocated the importance of flexible production lines and promoted internal combustion, hybrid and hydrogen-powered vehicles. Volkswagen subsidiary Audi is reducing its electric vehicle product line, while Volkswagen has shelved plans to sell a stake in its battery unit. A potential slowdown in electric vehicle

growth could jeopardize billions of dollars in industry investment and hamper compliance with emissions regulations. (RAUWALD, ECKL-DORNA, RAY, & Bloomberg, 2024) In conclusion, the litterature revealed that the regulation and policies of EV in Germany, challenges of German EV infrasturectur in the future, and German EV industry and current status of German EV market. Eventhough there are some chanllenges as Germany's current infrastructure is imperfect and the high price of electric vehicles makes some people hesitate to buy electric vehicles. The litteratures highlights the need for continued investment in infrastructure, supporting policies and innovative solutions to drive the transition to electric vehicles, as the German government and German automotive executives are also clear that electric vehicles are a big part of the future of the automotive industry. The trend is that German car companies currently have no technological advantage in the field of electric vehicle manufacturing. No longer an industry leader like internal combustion mechanisms.

2.5 Research gap and significance

The observation after reviewing the existing literature related to the research topic is that many research efforts focus on analyzing the policies surrounding electric vehicles in Germany, tracing the development of the German EV sector, and investigating the dynamics within the electric vehicle market, However, these viewpoints are typically discussed individually or paired together. For instance, the literature delve into the influence of Germany's EV policy on the development of electric vehicles, or the impact of Germany's EV infrastructure on the electric vehicle market. There is very limited of literature that integrates these three facets for a comprehensive analysis. Furthermore, there is even less literature that merges all of three aspects to study the strategic issues of multinational companies in Germany.

Thus, the research gap is defined since there is no current literature integrated with institution-based view, industry-based view and company's internal resources to analysis of BYD's business strategy in the German EV market. In addition, Germany has always been a country with a long automobile history which also occupies an important position in the automobile industry. BYD, as the largest manufacturer of electric vehicles. Thus, studying BYD's strategic development in the German EV market is not only of great significance to BYD, but also for other Chinese EV companies who seek expansion in Germany and has certain reference value for the entire electric vehicle industry.

Chapter 3 Theoretical Framework

This chapter presents the theories, concepts and models used in this study to answer the research question. This constructed by the Institution-based view, Industry based view focus on Porter's five force, Resource Based View with VRIO framework. The aim of this chapter is to enable user to gain a depth understand of framework and able to apply them in analysis afterwards.

3.1 Institution-Based View (IBV)

As globalization accelerates and multinational corporations expand, corporate management faces novel challenges, necessitating a global outlook to formulate effective strategies. The institutional landscape of economies and the significance of institutional evolution play crucial roles in the strategic management of multinational corporations. Adopting an institution-based perspective is pivotal for analyzing and discussing enterprise strategic development within the broader system.

According to Richard W. Scott, institutions are frameworks and practices shaped by rules, norms, and cognitive systems that uphold societal stability and imbue social behavior with meaning. (Scott, 2013) Building upon previous scholarship, Mike W. Peng refined the

institutional framework, delineating its scope in which encompasses both formal and informal systems governing individual and corporate behavior. (Meyer, Estrin, & Bhaumik, 2008) Formal systems encompass legal statutes, regulations, and similar structures, while informal institutions encompass norms, cultural values, ethics, and the like. These institutions are supported by three main pillars: regulative, normative, and cognitive.

The regulative pillar primarily involves the coercive authority of governments, serving as a cornerstone of the institutional framework. Regulation involves establishing rules, monitoring compliance, and implementing consequences or incentives to influence future actions. (Scott, 2013)

The normative pillar encompasses a prescriptive aspect that examines how the beliefs, values, and norms of relevant stakeholders influence the conduct of individuals and organizations. From a sociological standpoint, common values and beliefs are often prevalent within religious systems, social classes, business groups, and communities. (Scott, 2013) In his book "Institutions and Organizations: Ideas, Interests, and Identities," underscores the comparison between "appropriateness" and "instrumentality" as highlighted by scholars. When considering normative issues, the fundamental question shifts from "What choice will yield the best outcomes?" to "Given my role and the specific context, what behavior is most suitable for me to adopt?" (Scott, 2013)

The Cognitive Pillar, as defined by Scott revolves around the shared perceptions that define the essential characteristics of a social system and establish its interpretive frameworks. Scott emphasizes the "cultural-cognitive" label within his analytical framework to underscore how external cultural influences shape internal interpretive frameworks. Cultural systems have the capacity to influence cognition, permeate social values, and construct societal norms across various levels, ranging from the formation of social groups to the operational logics of organizations and the institutional patterns of national governance. (Scott, 2013) While

cultural-cognitive elements are widely recognized as consistent components in diverse systems, their degree of institutionalization often varies, leading to ongoing debates and negotiations during social reform and development processes. (DiMaggio & Powell, 1983)

160. Cognitive pillar aims to guide the behavior of individuals and enterprises toward internalizing the values and beliefs that are taken for granted, mainly focusing on informal institutions. With the existence of these three pillars, institutions can influence the behavior of individuals and enterprises. (Garrido, Gomez, & Juan, 2014)

	Regulative	**Normative**	**Cultural-cognitive**
Basis of compliance	Expedience	Social obligation	Taken-for-grantedness, shared understanding
Basis of order	Regulative rules	Binding expectations	Constitutive schema
Mechanisms	Coercive	Normative	Mimetic
Logic	Instrumentality	Appropriateness	Orthodoxy
Indicators	Rules Laws Sanctions	Certification Accreditation	Common beliefs Shared logics of action
Basis of legitimacy	Legally sanctioned	Morally governed	Comprehensible Recognizable Culturally supported

Table 1 - Scott's Three Pillars of Institutions (adapted from Koulikoff-Souviron and Harrison, 2008, 414)

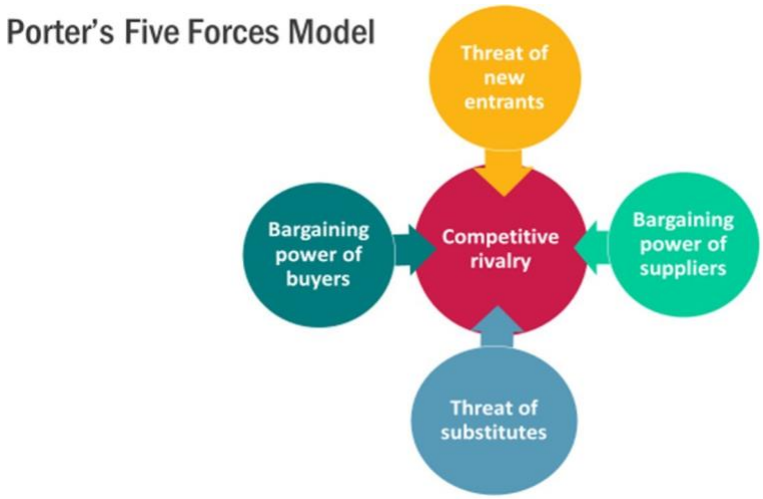
According to the IBV (Institution-based view), in order to devising a successful internaltional business strategy, it is significant to take the the formal and informal constraints of the specific institutional environment in which the business operates. it will not be sufficient to only analyze industry conditions and capabilities. Integrating IBV with perspectives from the industry and resource-based views (RBV) can provide a more solid foundation for strategy development. Institutions exert considerable influence on both the formulation and

implementation of strategy, as well as on the creation of competitive advantage. Hence, it's crucial to acknowledge that institutions are more than just contextual factors. (Peng, 2002)

3.2 Porter's Five Forces

Industry-Based View advocated by Michael Porter in his book"" Competitive Strategy" emphasizes the important impact of the industry environment on the enterprise" competitive advantage and performance. According to Porter, understanding the structural dynamics of an industry such as competitive forces, supply chains, technological changes enable companies understand the industry they operate and predict the trend of the industry development trends and competitor behavior thus helping company to develop corresponding competitive strategies. (Porter, Competitive Strategy-Techniques for Analyzing Industries and Competitors, 1998)

The Five Forces Model proposed by Porter as table in below to analyze industry's competitive landscape encompassing threat from competitors, potential new entrants to the market, the threat of substitutes, the bargaining power of suppliers, and the bargaining power of buyers.



(Figure 2 sources from Michael E. Porter, Harvard Business Review,1979)

3.2.1 Competitive rivalry

Interpreting thee forces accordantly, the threat from competitor usually discussing business competition. for example, the number of competitors within the industry, the more competitor the more competition in the industry. they all fight for scarps of market share. Moreover, the more similarities product or services they offer, the more intense competition they have, since the customer could easily switch. However, the competition usually less dramatic in an expending industry. The competition could also lead the price war between competitors, besides that, when look the competition between competitors, it could also look for exit barriers and fixed cost that Porter points out when industry with high fixed cost and high exit barriers, the company has strong temptation to cut price rather than cut production. (Porter M. , 1979)

3.2.2 Threat of new entrants

Potential New Entrant in an industry evaluate the entering barriers for an industry. General speaks, industry with lower entering barriers often have lower profit margin and firms have less market share. Factors as economic scale, production differentiation, capital requirement, access to distribution channel, regulation, switch cost are usually included when asses the new entrant in an industry. For instance, obtaining access to the distribution channel is crucial in the EV car industry. If the existing firm control distribution channels like retail stores or online platforms, new entrants would face the challenge of replicating such a structure while also competing with established firms on pricing, which is a complex task. Moreover, licenses, safety standards, and regulatory standards can create barriers, making market entry too cumbersome or expensive for new firms. (Porter M. , 1979)

3.2.3 Bargaining Power of Suppliers and buyer

Supply chain is powerful when a firm needs something important for their suppliers to stay in the business, when the power of supplier is high which may raise the costs otherwise limits the company resources. Factor as the number of suppliers, uniqueness which means supplier

provide the unique product, switch cost for switching supplier, forward integration in which means if supplier could move into the buy's industry that makes the former buyer difficult to compete. Do not forget the importance of industry as some industry or some sectors are interwind as the automobile, tech industry, the supplier need buyer to do well so they can do, thus there are normally balance power between the supplier and buyer. (Porter M. , 1979)

On the other hand, the five force also look the bargaining power of buyer as if customer is powerful that they could put pressure on business to provide better product or services perhaps in lower price as well. When looking the degree of buyer's power, those areas as the number of buyers, purchase size, switch costs, price sensitivity, and informed buyer may include. in practice, the fewer buyers, the more power buyer have such as in the aerospace industry, since there are fewer buyer in this industry, so buyer are powerful to demand favorable terms because the sellers depend on them.

3.2.4 Threat of Substitutes

According to Porter, threat of substitutes refers to when customer can find the substitutes in another industry that will threat companies in the former industry, take taxi as example, when uber and other ride-sharing appeared, it attracted customer quickly with its convenience and price advantage thus ride-sharing service industry competitor and threat to the taxi industry. Here are some ways to analyze the degree of threat of substitutes. The relative price performance that means when substitute performance is comparable or the cost of a substitute are low, the customer is easily switches to substitutes. Availability of close substitutes, if there are genuinely similar product on the market, the degree of threat of substitutes is high. Sometimes even though there are very different product, but customer still treat them as the same. (Porter M. , 1979)

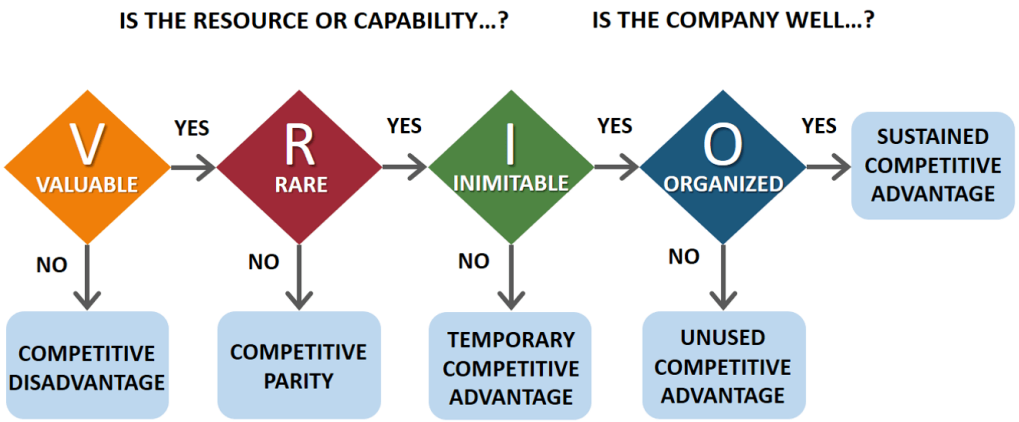
To sum up, by Porter's Five Forces, companies can gauge the attractiveness of their industry and adjust their competitive strategies accordingly to attain sustainable competitive advantage. Porter's Industry-Based View offers a framework to help companies thoroughly understand the competitive environment of their industry and make strategic plans and decisions to achieve enduring competitive edge.

3.3 Resource - based view, VRIO framework

Resource Based View is one of tradition perspective of strategy management focus on firm-level and describing and identifying the internal strength and weakness of the company and it is a managerial framework use to determine the strategic that can exploit to achieve competitive advantage. The roots of RBV found during the 90th, several scholars point it out. However, this paper focus on the interpterion of resource-based view introduced by Jay Barney in 1991. According to the definition of Barney, (Barney, 1991) resources include all assets, skills, technologies, organizational characteristics, and knowledge owned by the enterprise.

Resource-Based View emphasizes that a company's competitive advantage comes from its unique and difficult-to-imitate resource combination. These resources enable firms to provide value creation, scarcity, non-substitutability, and organizational capabilities to achieve sustained competitive advantage. RBV believes that corporate strategies should be dedicated to developing and utilizing these unique and valuable resources to create lasting competitive advantages. In addition, Barney also proposed two assumptions that underpin sustained competitive advantage. Firstly, resource heterogeneity suggests that firms within the same industry possess distinct bundles of resources, leading to varying levels of success. This implies that certain firms excel over others due to their unique resources. Secondly, resource immobility refers to resources that are challenging to trade or replicate. This assumption is also known as a VRIO framework that is a pivot measurement tool on whether resources can provide enterprises with sustainable competitive advantages. (Barney, 1991)

The resources could be divided to threshold capabilities and distinctive capabilities. Threshold resources refer to the basic resources and competencies required to enter and compete in an industry. The distinctive resource are those resources that can enable an organization to achieve competitive and the distinctive resources require ownership and control of resources and competencies that separate a business from the rest. VRIO refers to the four characteristics of Value (valuable), Rarity (rare), Inimitability (difficult to imitate) and Organized (organized). Only when resources or capabilities possess these four characteristics, they can consider as the sustained competitiveness advantage of an enterprise. (Barney, 1991)



(Figure 3 VRIO framework (Barney, 1991) , sources from the B2U)

Value, means the resource or capability must be able to help the enterprise exploit opportunities or solve problems, thereby creating value and provide value to the customer. If a resource or capability provides no value, it will not be a competitive advantage for the business. So, there are often question to consider, does the strength add value to your customer and can you capitalize on it to grow the Competitive advantage.

Varity is the second aspect of VRIO framework that underscores the necessity for a resource or capability to be relative scarce amongst your competitive set for an organization to sustain competitive advantage. This scarcity renders it challenging for other contenders to acquire the same resource or capability, thereby endowing you with a distinctive edge in your market. if these resources or capabilities becoming commonly and easily to access, it will no longer being an advantage that sustain the competitiveness. (Enochson, Onstrategy, 2022)

The subsequent step involves Inimitability which is to assess if the resource or capability is easily imitated or copied by other competitors. If it falls over time if the resource or capability can be easily imitated, it can not sustain a competitive advantage. If the strength creates value, is rare, and can not be copied, you could move to the final section of the VRIO analysis. however, If it's a strength, it's unique, it's rare, but can be copied, it may utilized in the short-term growth strategy.

Organized is the final step of the VRIO analysis that has less to do with evaluate the strength as earlier steps, is more about h the ability to effectively integrate, utilize, and manage resources and capabilities. if you can not leverage those resources, then it can not provide a competitive advantage and it is crucial for management team to consider for instance how are they going to leverage which resources, are they going to use it create the growth strategy, will they use which resource to drive their vision in the future and so on. if the management team know how to use this competitive advantage, they will all set. (Enochson, Onstrategy, 2022)

In summary, the VRIO framework helps a company determine its strength in its resources and capabilities from the firm-level service, VRIO framework serves to guide the corporation in developing its strategy to achieve competitive advantage.

3.4 Partial Conclusion

To sum up, the theoretical framework interprets three theories, which are institution-based view, industry-based view and resources-based view. Institution-based view emphasizes how

the institution in external environment of the enterprise affects the development strategy of the enterprise, such as regulations rules, norms, cultural or social cognition. Industry-based view focus on the Porter's Five Force: treat of new entry, bargaining the power of buyer and supplier, competitive rivals. Porter's Five force aim to plot the competitive landscape with an industry and understand the competitiveness of the particular enterprise in an industry and also the competition among its competitors. Finally, the theoretical framework interprets the resource -based view. from the perspective of the enterprise, the VRIO framework shows how the enterprise can find the advantages of resources and capabilities to help the enterprise to evaluate how the organization's resources contribute to the market position. These three theories focusing on strategic management that examine the dynamic relationship among institutions, organizations, and strategic choices provide comprehensive guidance for corporate strategic deployment as shown below

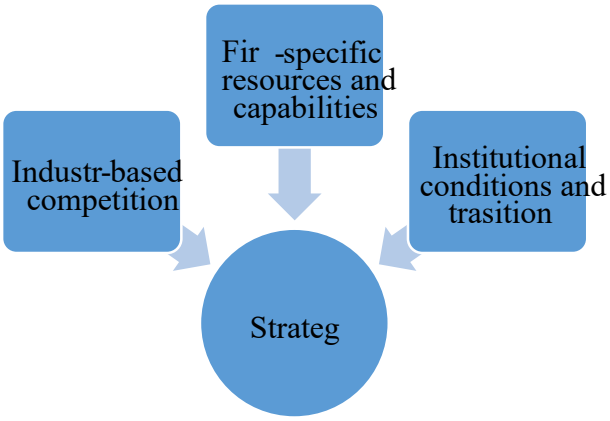


Figure 4: The Strategy Tripod, source: (Peng, Sun, Pinkham, & Hao, 2009)

3.5 Refection on the choice theories and models

When choosing the appropriate analysis method, I have thought about using the other two theories. For example, when doing macro analysis of the impact of external factors on corporate strategy, I have considered using PESTEL for analysis. PESTEL focuses on external environmental factors, including political, economic, social, technological, environmental, and legal aspects.PESTEL analysis and institutional theory both used in strategic management,

each offering a unique perspective on understanding business environments. PESTEL analysis provides a broader overview of external factors affecting organizations and institutional theory focuses on the impact of external conditions such as norms, rules, and regulations on corporate strategy. because for the EV car industry, its development relies heavily on the support of some external policies, Thus, in exploring how external factors influence BYD's strategy, this study opts for an institutional- based view over PESTEL analysis. (Aguilar, 1967)

This study also considered using SWOT analysis instead of Porter five force and resource based view, because SWOT analysis points out by Albert Humphrey stands for strength, weakness assesses, opportunities and threat and is a management tool to help company identify its internal strength and weakness, and external opportunities and threat in the market to have an understanding of company's current situation and how those factors affect company business.

But SWOT analysis may oversimplify complex strategic issues and fail to prioritize or weigh factors appropriately, analysis of industry and company resources. The analysis is also not comprehensive. SWOT analysis does not provide guidance on how to address identified issues or develop strategies beyond highlighting them. (Helms & Nixon , 2010)Therefore, industry-based view focus on Porter's Five Force and resource-based view using VRIO framework to answer What opportunities and challenges does BYD face in the German EV market? And which resources can BYD exploit to achieve a competitive advantage in the German EV market?

While make this decision, the limitations of Porter's Five Forces and the VRIO framework have been considered. Porter's model fails to account for important factors such as the dynamic interactions among competitive forces, the diverse behaviors of competitors, the evolving nature of industry boundaries, and the role of innovation, differentiation, and branding in shaping competitive advantage. Similarly, the VRIO framework's static nature hampers its ability to respond to changes in the competitive landscape, advancements in

technology, and shifting customer preferences over time. Additionally, it tends to overly prioritize the rarity and uniqueness of resources, potentially overlooking factors such as resource substitutability and the coherence of organizational strategies needed for long-term competitiveness.

Despite these limitations, the combination of these theories as mentioned in (Figure 4) co-establish the comprehensive analytical response to business strategy, thus it is the best choice for this study.

Chapter 4 Methodology

This chapter introduces the methodology that is applied to this research to answer the research question: *How can BYD adjust its business strategy adapting to the German EV market to increase sales?* Th research design is guided by the "Research Onion Model" from Saunders. (Saunders, Lewis, & Thornhill, 2012) The purpose of applied this model is to clarify the methodology path step by step that is practical and convenient for research to construct the research design.

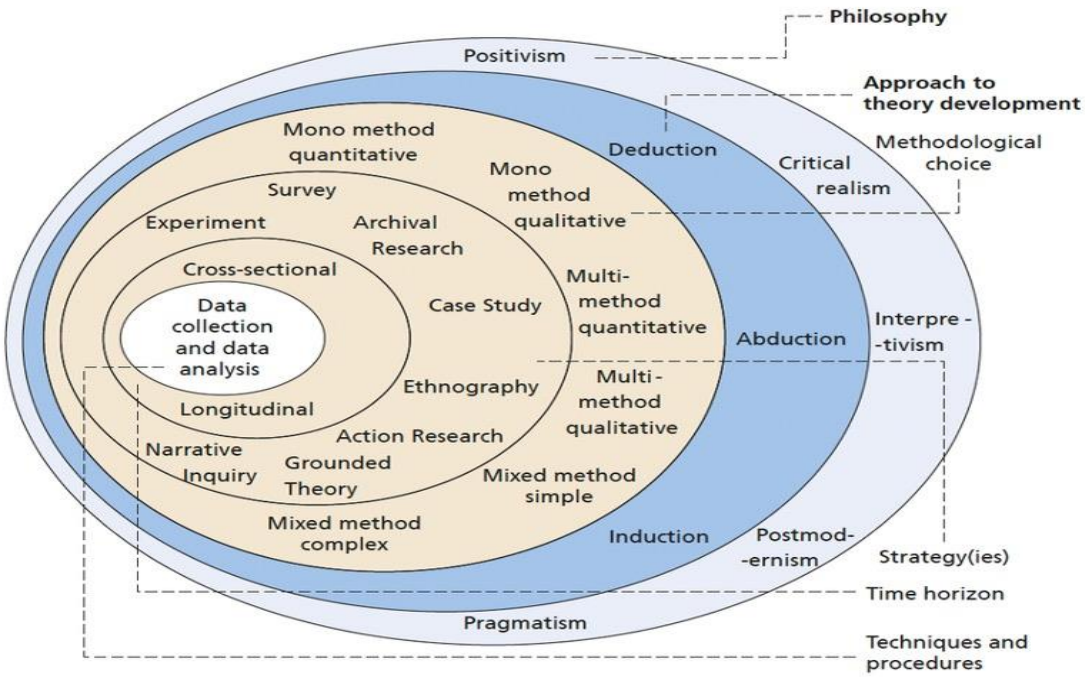


Figure 5: The "Research Onion" Model from Saunders, Lewis,&Thornhill, Research model for Business Students

4.1 Research Philosophy -Pragmatism

According to the Research Onion Model, research philosophy is the out layer of the model which should be decide firstly. It is necessary to explain the underlying research philosophy because a research design is built on the appropriate research philosophy. Different philosophical assumptions directly affect the research design and the quality of the design. (Easterby-Smith, 2015)

The choice of research philosophy is based on the research's perception in term of ontology, epistemology and axiology. Ontology refers to the assumptions about the reality, to determine the choice of otology, one must answer" what is the nature of reality?", epistemology refers to the assumption about the knowledge, people likely ask "how do we know what we claim to know?" in order to establish the choice of epistemology. Axiology concerns about the values that guide our research to begin with.

by consideration the question above and the overall aim of this research, the pragmatism research philosophy is selected because pragmaticism philosophy allows research to fully engaged in method, complexity and richness of world and focuses on the usefulness and applicability of research findings, rather than an all or nothing philosophical position compared to positivism and interpretivism. Moreover, pragmatism also provide research the great flexibility in terms of method for designing and data gathering. (Saunders, Lewis, & Thornhill, 2012)

4.2 Research Approach - Abductive

The second layering of Research Onion model is approach to the theory development.

An abductive approach is adopted in this study. This involves inferring new theories and propositions and "seeks a situational fit between observed facts and rules" (Timmermans & Tavory, 2012) In order to demonstrate a full picture of the topic of interest with an emphasis on the unique implications of the case of BYD. In the process of generating research questions, I employed systematic combining to align theory with reality as well as constantly redirecting

my research. Initially, I identified potential areas for investigation based on various interests. With a preliminary topic in mind and armed with relevant theories, I went through my first round of data review, coding information from literature reviews into categories and looking for common themes. An open mindset was crucial through the data combining process in order to constructing the research framework organically, ensuring the validity and relevance of the research questions. As I arrived at a more specific preliminary research question, I were able to redirect further research in a top-down fashion where I actively searched for data relevant to the question of interest. This non-linear approach enable me to go back and forth to successively modify the research to incorporate findings made along the way. (Dubois & Gadde, 2002)

4.3 Methodological choice – mix-method

The third layer from the "Research Onion Model" is methodological choice. In order to meet up the aim of this research, the mix-method is adopted. The mixed- method allows the research to combine the quantitative research and qualitative research that could help me create more complete picture to answer the research question. The numerical data used to find the pattern and make assumption while the non-numerical data such as text, document and the video would help me to understand the concept, opinion and to gain the in-depth insight to my research question. (Saunders, Lewis, & Thornhill, 2012)

4.4 Research strategy – singular case study

The 4th layer of "Research Onion Model" concerns the research strategy. This study chose to adopt a case study research approach as research strategy. The case study approach focusing on specific phenomena of organizations or individuals in real settings. (Easterby, Thorpe, R, & Jackson, 2015) As George and McKeown (George & McKeown, 1985) pointed out that a major advantage of the qualitative case method is that it can address the effects of organizationally active phenomena in real-world contexts. Through a case study approach, key factors and causal relationships can be more easily discovered and tested.

The rationale for selecting BYD as the sole case company in this study aligns with the argument put forth by Dubois and Gadde (2002) as referenced by Towers and Chen (2008). They posit that in situations involving numerous independent variables and intricate structures, delving deeply into the analysis of a single case is more beneficial than expanding the sample size for observations. Given BYD's status as the world's largest manufacturing company in electric vehicles, studying it intensively can yield significant insights that are transferrable to other companies within the industry. This focused approach allows for a comprehensive understanding of key phenomena, benefiting not only BYD but also contributing valuable reference points for others in the same sector.

4.5 Time horizon and Data Collection

The fifth layer of " Research Onion Model" is time horizon, there are two type of time horizon which are cross-sectional and longitudinal approach. The cross-sectional approach often involves case studying at a particular time and study the occurrence of a phenomenon (Saunders, Lewis, & Thornhill, 2012)while longitudinal approach often involve secondary data has been used to gain insight into historical developments of the industry or the company. Based understanding of difference of two type of time horizon, the cross-sectional approach is used in this study since it is shorter, however, the longitudinal approach is used when investigate development of the case company through the secondary data. research scope and the research purpose of a study are the most significant factors leading the researcher to select research methods, which concern the particular research tools and procedures to conduct research and analysis. (Yin, 2011)

4.6 Data Collection and Data Analysis

The last layer of " Research Onion model" refers to the data collection and data analysis. Both qualitative and quantitative data originating fom various sources should be utilized to do the research. (Wahyuni, 2012) During the data collection process in a case study, the high complicacy and diversity of data sources may bring impediments to the research process.

It is essential for researchers to be capable of dealing with this complexity by distinguishing the most relevant and useful data according to the basis of research scope. (Yin, The case study as a serious research strategy. Knowledge, 1981) In this study, the secondary sources account for the major part to investigate and answer the research question.

For the sake of heightening the validity of secondary data, I found that validating the credibility and timeliness of the data's source is fundamental. Comparing and examine the reliability of the data materials from diversified channels increase the validity of results. In the choice of the data, the secondary data used in this study comes from reliable information and data such as the company's official website, Statista, the government's official website, and renown journals because of the completeness and authoritativeness.

Main sources of secondary data in this study are as follows:

- German EV government and institutions, subsidies from United Nation website and academic literature
- Theories and Models from academic literature and journals
- Online video materials of relevant interviews, speeches
- Insight from industry journal
- Reports from BYD official websites and other company websites
- Industry Insight from Consulting firm

Concerning data analysis, secondary quantitative data such as statistics were used primary to understand the German EV market while secondary qualitative such as literatures, content analysis, videos etc. aim to provides information related to the research question. The data analysis provides the comprehensive information that enable researcher to understand and the factor, cause, trend from different sources and it is a foundation for the findings in this study.

4.7 The limitation of the Methodology

This study as mentioned used secondary data as mentioned above, which means the primary data will not be collect and analysis. It has been aware there could have few disadvantages of only using the secondary data, firstly the secondary data may lack of relevance which means it is may be difficulty for the researcher to find the data that exactly match to answer the research question, not as the primary data which research have more control over the design of the data, e.g. researcher could design interview questions customized to answering the search question. In contrast, an interview from the secondary source may contain other information which irrelevance for the research purpose. In addition, not as the primary data, the researcher may have less control of data quality of secondary data when the secondary data coming from complex sources, it is difficult for researcher to go through all the sources to evaluate their quality.

However, this study has recognized the limitations of using secondary data. In order to cover these limitations, this study extensively collected different types of data when selecting secondary data to answer the same question， ensuring the integrity of the information. In terms of data extraction, this study retains data related to research questions through screening, ensuring the relevance of the data. Finally, considering data quality, as mentioned from data validation, the secondary data used in this study are from some large official institutions, company websites, or well-known newspapers or academic forums.

Chapter 5 Description Section

This chapter provide background knowledge relate to the research top of case company will be present here. It includes the development of BYD, BYD's business strategy as well as the

current sales model of BYD's EVs in the German market. These background knowledges sever a solid foundation for the analysis.

5.1 Background of BYD development

BYD founded in 1995, in Shenzhen,China. BYD is a large Chinese listed automobile company with business in electronics, automotive, new energy, and rail transit. BDY's market value exceeds 13 billion us dollars and has more than 220,000 employees worldwide. According to annual financial report of BYD, there are 3.02 million units Electric vehicles sold in 2023, the total revenue of 602.315 billion yuan with a year-on-year increase of 42.04% . Its net profit reached 30.04 billion yuan, a year-on-year increase of 80.72%.

demonstrating an impressive year-on-year growth of 80.72%. Central to BYD's success is its development philosophy, which emphasizes "technology is king, innovation is the basis." Leveraging its strong research and development capabilities and innovative business model, BYD has made significant advancements in various sectors, including batteries, electronics, passenger vehicles, commercial vehicles, and rail transportation, solidifying its position as a prominent player in the industry. (BYD, BYD periodic reports, 2023)

BYD demonstrates remarkable performance across all its business sectors. In the battery realm, BYD boasts over two decades of relentless innovation, resulting in 100% independent capabilities in research, development, design, and production. The company has established a complete battery industry chain, offering a range of products including consumer 3C batteries, power batteries (such as lithium iron phosphate and ternary batteries), solar cells, and energy storage batteries. Notably, BYD stands as the world's largest manufacturer of lithium iron phosphate batteries, with its products distributed globally.

In the electronics domain, BYD maintains its position as an industry leader, leveraging cutting-edge technology and large-scale production. Its clientele includes prominent players in the global consumer electronics market, such as Samsung, Huawei, Lenovo, Apple, and

Microsoft. In automotive innovation, BYD focuses on research and development, particularly in core areas like batteries, motors, and electronic controls for new energy vehicles. Adhering to the 7+4 strategy, BYD aims to achieve comprehensive coverage of new energy vehicles in both passenger and commercial sectors, spanning private cars, taxis, urban buses, road passenger transportation, urban logistics, and specialized vehicles for warehousing, ports, airports, and mines. The BYD K9 electric bus, widely utilized in numerous cities worldwide, exemplifies the company's success in the commercial vehicle sector. (Book, et al., 2024)

In the passenger car market, BYD has emerged as a leader in China's electric vehicle industry, offering models with impressive acceleration, all-electric 4-wheel drive safety, and exceptional fuel efficiency. By attracting international talent, including renowned figures like Wolfgang Egger and Heinz Keck, BYD Auto is advancing towards an international strategic direction. Moreover, BYD has invested significantly in research and development, assembling a team of over 1,000 experts and allocating substantial funds to develop an urban transportation cloud rail system. Compared to traditional subways, this system offers cost-effectiveness, shorter construction periods, reduced noise levels, and adaptability to diverse urban terrains, positioning it as a promising future urban transportation solution.

In August 2022, BYD officially announced its entry into the German market and cooperated with Hedin Mobility, a leading European dealer group, to provide high-quality EV products for the German markets. The main passenger car models currently launched in the German market include the BYD Tang and Han series, which are priced between 50,000 and 70,000 euros, and the BYD seal and atto3, which are priced between 30,000 and 50,000 euros, and

will just enter the German market in 2024. The price of BYD's compact model BYD Dolphin is less than 35,000 euros.The article “An Existential Crisis in Germany Auto industry ” from the Spiegel international points out that Management company Berylls and opinion research company Civey recently conducted a survey among 1,400 high-end German car owners, and

found that nearly 61% of them are open to buying Chinese cars. Especially among car owners aged 30 to 39, the main consumer group in the next few years, the response is better. German consumers' acceptance of BYD has also increased significantly. Judging from the picture, from October 2023 to February 2024, it increased from 39% to 50%, and the acceptance increased by 11%. (Book, et al., 2024)

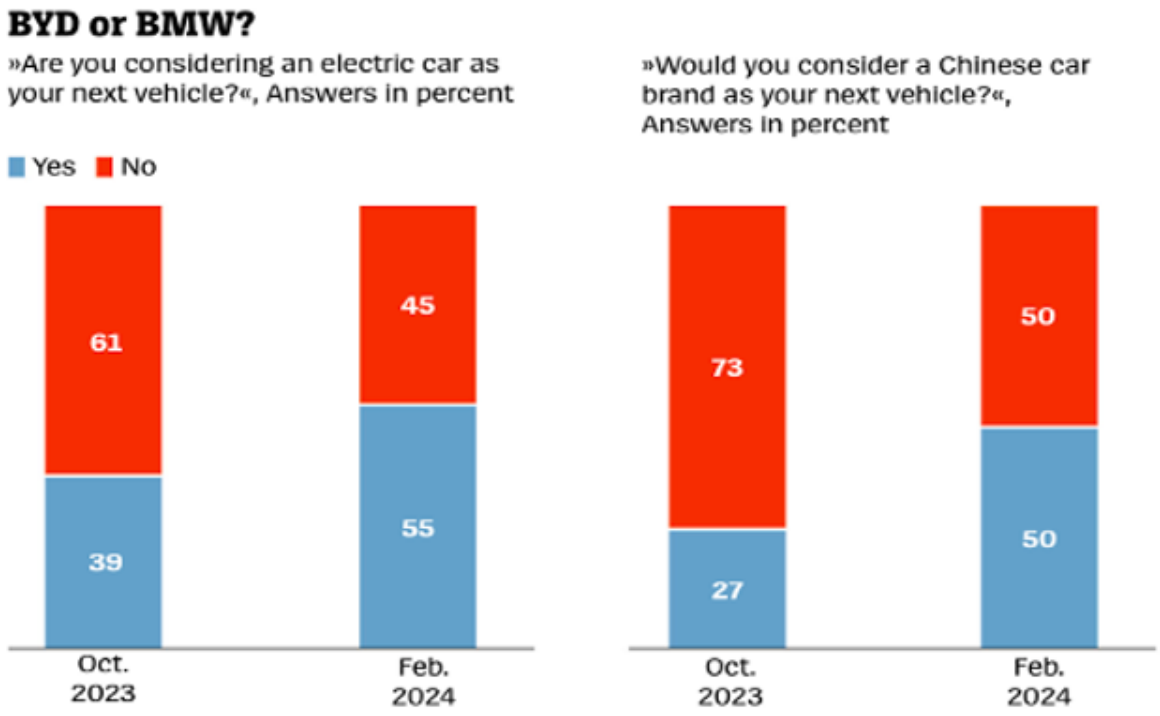


Figure 6 "Carwow survey"of consumer preference, sources from Der SPIEGEL. (Book, et al., 2024)

Despite the growing dominance of Chinese EV manufacturers like BYD in Europe, there is discontent within the E-mobility sector regarding the selection of a non-European firm for such a prominent event. European manufacturers are still in the early stages of electrification, while their Asian

counterparts have been deeply involved in the transition for years. BYD's sponsorship with UEFA European Football Championship 2024 highlights the increasing influence of Chinese EVs in Europe, fueled by competitive pricing and advanced technology. However, questions

arise about why a local brand was not chosen and whether price is the sole determining factor in the market. To strengthen the national EV market and promote sustainable mobility, industry experts suggest additional measures such as supporting charging infrastructure expansion, incentivizing EV purchases, and promoting public transportation services. (Platin, 2024)

5.2 BYD passenger car in Germany

As a global leader in EV manufacturing, BYD has consistently dominated China's EV market for eleven consecutive years. Despite facing formidable competition, the company has experienced rapid business expansion, underscoring its industry expertise. Committed to strengthening its international presence, BYD has prioritized entering global markets.

In 2022, BYD announced a strategic partnership with the European dealer group Hedin Mobility to introduce its EV to Germany, marking its debut in the European passenger car market. Germany holds significant importance in BYD's European expansion strategy. Renowned as the birthplace of the global automotive industry and home to Europe's largest automobile market, Germany boasts stringent automotive standards and a discerning consumer base. (Daily, 2022)

BYD's entry into the German market presents an opportunity to showcase its products and technologies in one of the world's most competitive markets. This strategic move is poised to enhance BYD's brand reputation, market positioning, and global presence, aims to accelerate innovation and development in the EV industry, contributing to the advancement and prosperity of EV industry globally. In September 2023, BYD will debut at the Munich Auto Show with 6 models. These 6 models include BYD HAN, BYD Seal, Song PLUS, BYD ATTO 3, BYD Dolphin, and D9 debuted at IAA MOBILITY in Munich, at the same time

BYD announced that the Seal was officially launched in Europe. The major sales models of BYD's EV in Germany include BYD TANG, BYD HAN, BYD SEAL, BYD ATTO3, and BYD Dolphin. (BYD, BYD, 2023)

BYD HAN: Highest selling price in Germany. the sale price is 69020 various from 69020 euro to 72590 euro depends on models. Equipped with 85.4kWh blade electricity, accelerating to 100 km/h in 3.9 seconds. WLTP battery life from 521 kilometers combined to 662 kilometers in the city.

Fast charging reaching 80% in 30 minutes. Winner of the iF Design Award in 2021, marking the first Chinese car brand to receive this honor. (BYD, BYD EV model, 2024)

BYD TANG: Priced at €69,000, accelerates to 100 km/h in 4.6 seconds with 380 kW motor power. WLTP battery life from 400 up to 528 kilometers in the city, zero-emission electric driving mode. (BYD, BYD EV model, 2024)

BYD SEAL, priced from €44,990 to €50,990, with WLTP range up to 700 kilometers. Fast charging reaching 30-80% in 30 minutes. Luxurious design and performance options, with acceleration to 100 km/h in 3.8 seconds for AWD version. (BYD, BYD models, 2024)

BYD ATTO 3: Cost-effective compact SUV, priced from €37,000 to €39,000. WLTP range up to 568 kilometers in city driving. Spacious trunk capacity of 440 liters, expandable to 1338 liters with rear seats folded down. Recognized as a finalist for "Family Car of the Year 2023" award, praised for design and practicality. (BYD, BYD EV model, 2024)

BYD Dolphin: small electric car, priced at €32,990.WLTP endurance of 427 kilometers, acceleration from 0-100 km/h in 7 seconds.Spacious trunk volume of 1310 liters when rear seats are folded.Highly praised for advanced electric car technology
and competitive pricing, challenging European competitors. (BYD, BYD models, 2024)

5.3 BYD's business strategy

5.3.1 "7+4" Strategy

BYD's introduced the "7+4" strategy in 2015. The objective is to achieve a full industry chain layout, improve the company's competitiveness and sustainability while better adapting to market changes and demand shifts. The "7" in the strategy signifies the company's concentration on various automotive sectors, encompassing commercial and passenger vehicles, public transportation, light and heavy logistics, car rental, and financial services. BYD's implementation of the 7+4 model focuses on several key areas:

- Diversified business: with a diversified business model, it helps BYD to diversify the risks and to better respond to market changes and demand shifts.
- Technology leadership: By entering emerging fields, it forces BYD focus on the technological innovation that helps BYD continuously improve its technical capability to obtain certain technical advantages and support and guarantee the future development.
- Improve brand awareness: Operating under a diversified business model, BYD increases its opportunities for business growth by catering to customers in different markets and meeting various needs., which can further enhance brand awareness and market share.

(BYD, 7+4 full market strategy, 2022)

5.3.2 BYD's Integrated Vehicle Intelligence Strategy

As artificial intelligence (AI) continues to advance rapidly, intelligent driving and smart systems are becoming pivotal trends that are shaping the trajectory of the electric vehicle (EV)

industry. In response to this, BYD unveiled its “Integrated Vehicle Intelligence Strategy “ in 2024, aiming to lead the advancement of intelligent automobiles and accelerate the transformation of the automotive sector towards intelligence. This strategic initiative seeks to improve user-friendly, enable more personalized settings, and meet diverse user needs.

The "Xuanji Architecture," is the first intelligent-electric fusion architecture, and it is an essential component of vehicle intelligence. It provides the car with an intelligent "brain" and "neural network", enabling it to fully perceive, contemplate, control precisely, and execute collaboratively like advanced intelligent life forms. This architecture enables real-time capture of changes in internal and external environments, swiftly consolidating information feedback to a central "brain" for decision-making, quickly adjusting the "body" vehicle status, and greatly improving driving safety and comfort. The Xuanji Architecture seamlessly integrates the E-platform with intelligent parking technology. Combined with the flexible and smooth control capabilities of the four motors, it can automatically achieve precise rotation and parking, effortlessly maneuvering into even the most challenging spots.

Meanwhile, BYD pioneered the first dual-cycle multi-modal AI large vehicle. boasting the industry's largest database, leading sample size, and high computing power, covering more than 300 vehicle scenarios, endowing vehicles with the ability to continue intelligent evolution. Furthermore, a $5 billion investment will be made to construct the world's first batch of global all-terrain professional racing venues. In collaboration with DJI, is creating the world's first integrated vehicle-mounted drone, pioneering an integrated vehicle-grade drone hangar supporting intelligent storage, automatic battery exchange, and charging management. (BYD, BYD Showcased the Intelligence Advancement at Dream Day 2024, 2024)

Chapter 5 Analysis

In order answering the main research question: *How can BYD adjust its business strategy adapting to the German EV market to increase sales?* the analysis takes focus on investigated sub three sub-question. The first sub-question, examining the institutional opportunities and challenges faced by BYD in Germany, will be explored through the lens of the institution-based view. The second sub-question, analyzing the opportunities and challenges encountered by BYD in the German EV market, will be assessed using Porter's Five Forces framework. Lastly, the third sub-question, identifying the resources that BYD can leverage to achieve a competitive advantage in the German EV market, will be illustrated through the application of the VRIO framework. The analysis section serves as the central component for generating the findings.

5.1 Institutional opportunities and challenges BYD face in the German EV market?

Being the nation with the most extensive legacy in the automotive sector, Germany is actively fostering the advancement of electric vehicles amidst the global automotive industry's shift towards sustainability. In pursuit of this objective, the German government has devised a range of policies and incentives aimed at propelling the adoption of electric vehicles. As mentioned in theoretic view, institutions has been defined by different scholars, W. Richard Scott, theorized the conception of institutions in which Institutions consist of regulative, normative, and cultural-cognitive components, which, along with related activities and resources. When institutions distinguish and constrain behaviors through legal, moral, and cultural constraints, it is equally necessary to realize that institutions can also empower and support social actions through incentives, guidance, and resources. (Scott, 2013)

5.2 BYD's opportunity and challenge of German EV industry from perspective of Porter's five force

As discussed before, the institutional- based view offers the insights into the competitive nature of the industry and understand impact of the industry environment on enterprise, this section will use Porter's five Force help BYD to understand the structural dynamics of the German EV industry in term of the threat of new entrants and subsidies, the bargaining power of supplier and buyer and the internal rivalry in order to understand industry development trends and competitor behavior, thus helping BYD to identify opportunities and threats in the German EV industry in order to develop corresponding competitive strategies.

5.1 Institutional opportunities and challenges BYD face in the German EV market?

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From Scott's point of view, the existing policies for the German electric vehicle industry are mainly formal institutions such as regulations and some government incentive policies. the German federal government formulated the Climate Action Plan 2050 in 2016 to ensure that Germany achieves the targets set by the Paris Agreement and the EU's climate goals. This document underscores the importance of promoting EV and proposes various measures to accelerate their adoption. These include providing purchase subsidies, building charging infrastructure, and reducing the cost of electric vehicle usage. For instance, tax incentives

outlined in the literature review include a ten-year ownership tax exemption for battery electric vehicles (BEVs) and fuel cell electric vehicles (FCEVs) for individuals, as well as corporate tax incentives such as a reduction in taxable amounts for BEVs and plug -in hybrid electric vehicles (PHEVs (Ministry, 2016)

The personal purchase subsidy previously amounted to 4,500 euros that has been canceled the subsidy at end of 2023. Currently, there is no government subsidy for the personal purchase of EVs. The sudden termination of the subsidy has made many consumers and businesses very distressed. In addition, in term investment of charging infrastructure, according to the he "Climate Action Plan 2050" that German government set target for installing charging infrastructure in key areas such as cities and highways, accompanied by financial support. This emphasizes the importance of charging infrastructure construction to support the development and popularization of EVs In 2022, German government approved a 6.3 billion euro investment plan spanning three years to rapidly expand the nationwide electric vehicle charging network. This support provides a vital foundation for BYD's operations in Germany and facilitates the sale of electric vehicles in the country. (Ministry, 2016)

However, there are also some issues that need to be paid attention to in BYD's strategic development. For example, the biggest problem currently is the serious shortage of charging piles in Germany, and the construction of charging piles is lower than expected. As mentioned in the literature review, There is still a long way to go to achieve the previous strategic goal of 1 million charging stations by 2023 set by German Transport Minister Wiesing. To achieve the goal, Germany must double its current rate of infrastructure development. Moreover, the distribution is uneven, with the average distribution in big cities being significantly more than in remote areas and rural areas, which has affected the sales of electric vehicles in towns and cities with insufficient charging piles. This has brought certain obstacles to the promotion and popularization of electric vehicles.

In summary, it's evident that Germany is deeply committed to developing the electric vehicle industry. Many regulations and incentive policies for electric vehicles are made by German government to promote the development of the electric vehicle. However, some incentive adjustments were made at the end of 2023, which could have a negative impact on the EV sales.

5.2.1 Threat of New Entrants

Entering the electric vehicle industry requires massive capital investment, massive R&D expenditures, industry experience, and economies of scale to achieve efficient manufacturing. For instance, BYD's R&D expenses was 1.105 billion dollars increasing to 5.05 billion dollars for from 2019 to 2023. (finbox, 2020)

Other EV manufacturers have also invested large sums of money in R&D investment, Tesla's R&D expense was 94 million dollars in 2010 and increase to almost to 40 million US dollars in 2023, (statista, 2024) In addition, several other well-known German car brands have also continued to invest a lot of money in their EV development, In early 2024, the Volkswagen Group has committed to investing 122 billion euros on electrification and digitalization over the next five years. (Volkswagen, 2023)

Daimler AG's Supervisory Board has approved a 60 billion euro business plan to Mercedes-Benz Business for 2022 to 2026 in order to keep high level of R&D expenditure for the electrification of the product portfolio and digitalization measures and automated driving. (Group, 2021)

BMW's R&D investment in EV has also exceeded expectations, According to info from the Bloomberg, the R&D expenditure in 2023 Q3 was 3,396 million euro which has an year on year increasing of 15,4%, the R&D expenditure went mainly into the development of the new

models, electrification, digitalization of the vehicle fleet, and automation driving. (BMW, 2023)

In addition, the EV industry is unlikely to achieve particularly large profits in a short period of time, so the initial sunk costs are quite large. As we all know, cars are one of the most complex industrial products, and the production of a car requires hundreds of parts and the battery of EV, the most critical component of electric vehicles, accounts for almost one-third of total manufacturing cost. Overall, the cost of electric vehicles is very high. In the first half of 2023, the prices of cobalt, lithium, and nickel increased by 50%, 100%, and 30% respectively, which directly led to the increase in the price of power batteries. Since the new car-making forces do not have an independent battery supply chain, high-speed train batteries are still supplied by external suppliers, thus bearing the cost pressure of rising battery prices. For example, the emerging Chinese electric vehicle brands NIO and Xpeng which are also on sale in Germany. (St, 2024)

In the first half of 2023, NIO's total revenue was 20.23 billion yuan, a year-on-year increase of 22.96%; but the net loss was 4.54 billion yuan, a year-on-year increase of 369.6%. Based on the delivery volume of 54,600 vehicles, Weilai lost 83,100 yuan for each car sold. The situation of Xpeng Motors is not optimistic either. In the first half of the year, its revenue was 14.891 billion yuan, a year-on-year increase of 121.85%; but its net loss was 4.402 billion yuan, a year-on-year increase of 122.21%. Based on the delivery volume of 44,100 vehicles, Xiaopeng lost 99,800 yuan for each vehicle sold. (St, 2024)

Thus, the entry barriers of EV battery is high of its large initial sunk cost of the electric industry is and the current low profitability. Looking at the global automobile industry has a history of hundreds of years, the well-known automobile brands are still the ones that most people is familiar with which means not every company has capability to entering the EV industry, especially for emerging brands which do not have the same resources and expertise to develop as established automobile manufacturers, the infrastructure to produce electric vehicles and establishing a foothold in the industry comes at a huge cost.

5.2.2 Threat of Substitutes

When we discuss alternatives to the electric vehicle industry, first of all we realize that the development of electric vehicles is to reduce carbon dioxide emissions and contribute to improving the environment. Many people believe that electric vehicles should not be the only solution, as mentioned in the literature view, in addition to electric vehicles, there are also hydrogen energy vehicles on the market.

Fuel cell electric vehicles (FCEVs), also known as hydrogen cars, operate by converting hydrogen directly into electrical energy, producing only water vapor and warm air as emissions. However, compared to the electric vehicle (EV) industry, FCEVs and their supporting hydrogen infrastructure are in early stages of development. Literature reviews indicate minimal FCEV sales in the German market, with only 263 FCEVs registered in 2023, a significant drop from previous years, in contrast to the 1.4 million registered BEVs. Although FCEVs offer fast refueling, challenges persist, notably safety concerns regarding hydrogen storage and fuel cell operation. Hydrogen is highly flammable and improper handling may lead to explosive accidents, while fuel cell devices pose risks of leakage or overheating if not properly designed or manufactured (Hydrogeninsight, 2024).

Furthermore, the inadequate infrastructure presents a significant barrier to FCEV adoption, with only 88 hydrogen refueling stations in Germany as of 2023. Analysts attribute the global failure of FCEVs to take off since 2014 to factors such as insufficient hydrogen refueling infrastructure, high hydrogen costs, and the upfront expense of FCEVs (Observatory, 2024).Currently, German taxpayers are also collectively calling on the government to stop the substitutes for the FCEV since the German government spent nearly 500 million euros on hydrogen vehicle subsidies in 2007.

The national taxpayers' Bund der Steuerzahler (BdSt) described this as "absurd" and “pointless” as, they mean that hydrogen cars are much less efficient than battery-powered electric cars, and a recent analysis by DSi, the group’s research arm, also noted that “at least in some areas, the use of hydrogen. There are technical issues. This certainly includes the passenger car industry." And using green hydrogen made from renewable energy is a "waste of resources" because of the waste when converting clean electricity into hydrogen and back again inside a fuel cell vehicle. causing energy loss. (Collins, 2023)

The analysis shows that although electric vehicles are not the only solution in terms of environmental protection and emission reduction, cars can also use environmentally friendly fuels, such as the hydrogen energy vehicles (FCEV) mentioned. However, it is evident from the analysis that FCEV are not a better solution than electric vehicles (EVs), and in the case of limited funds, it has led to limitations in the development of hydrogen energy vehicles. From the perspective of substitutes, challenging EVs necessitates presenting a superior alternative. Presently, there is no other solution on the market that has lower investment and higher safety than the electric vehicle in renewable energy sources. Therefore, EVs will continue to dominate as the primary choice for environmentally friendly and emission-reducing vehicles for the foreseeable future.

5.2.3 Bargaining Power of Suppliers

Automobile manufacturing involves hundreds of parts, and suppliers run through the entire value chain, Suppliers have a crucial role in the EV industry regardless in which countries. The core components of EVs include motors and batteries, electronic control, chassis, body, central control system, interior, exterior decoration and other components. In Appendix 1 demostrates BYD's EV core components includes the power system , electronic control central control system,body chassis. in which most of components are made by BYD itself. It is evident that a vertically integrated company can manage the entire production process.

Batteries are about one-third of the cost of the entire vehicle, scaling and in-house production of batteries is important to reduce overall production costs and provide attractive products. Outsourcing limits access to expertise and increases pressure on EV manufacturers. Except for BYD, many others including Tesla, Germany's Volkswagen, BMW, etc. do not have electric vehicle battery technology and still rely on external suppliers, so key components of electric vehicles like batteries The bargaining power of suppliers is relatively high. (Pistilli, 2024)

In addition, battery prices have increased significantly in recent years due to raw material issues. The reason is that the expansion of battery production takes about 9 months to a year to complete, but the expansion of lithium mines requires 15 to 24. The main source of battery expansion is Due to sufficient capital chain, and lithium ore and other bulk products, the expansion of production capacity is slow. Firstly, it is affected by the time required to expand mineral production capacity, and secondly, it is also limited by the state's constraints on the production capacity specified in the company's mining license. Moreover, cobalt and nickel are important raw materials for ternary lithium batteries, and the market demand is also very tight. This is the reason for the increase in raw material prices. It shows that battery suppliers generally have low bargaining power on upstream raw materials. This has a great impact on the most downstream EV companies and increases the manufacturing costs of electric vehicles. (Peters, 2024)Thus, because battery suppliers are the core of the electric vehicle industry, and many electric vehicle companies do not have the technology to produce batteries, they all rely on supplier supply, that indicate the price bargaining of core component suppliers in the EV industry is high.

Although this is not good for the industry as a whole, it is an opportunity for BYD. BYD CEO Wang Chuanfu emphasized in an interview that the company has independent research and development capabilities for almost all core components, including car tires and glass. Fodi Battery, a wholly-owned subsidiary of BYD, provides batteries for BYD's electric vehicle manufacturing. BYD signed a mining strategic cooperation agreement with the Nanchang

Municipal Government in China, which is an important milestone. As part of the agreement, BYD plans to invest 28.5 billion yuan (approximately US$3.95 billion) to establish a comprehensive development and utilization production base in Yichun City. Subsequently, BYD Yichun Mining Co., Ltd. was established. The company's business scope includes the mining of non-coal mine mineral resources and the sales of non-metallic minerals and products. These wholly-owned subsidiaries enable BYD to achieve self-sufficiency in battery raw materials. With such a volume of ore, BYD could lock in lithium needs for more than a decade, the report said. The mines are expected to produce lithium carbonate at a cost of less than 200,000 Chinese yuan per tonne, far below the domestic price of 500,000 yuan per tonne in mid-March, the report said.Byd's integrated supply chain allows BYD to adjust its manufacturing costs, giving BYD a very large cost advantage compared to other electric vehicle companies that require external supply of batteries. (Xue, 2022)

5.2.4 Bargaining Power of Buyers

Barging power of buyer represents the ability to negotiate the price and conditions with the EV company, In the German EV Industry, the individual buyer normally purchase one car at a time, in this case the bargaining of buyer is low compares with if it is a bulk purchase, the bargaining power of buyer will be higher than the individual car purchases. As mentioned in the literature review, Germany currently has just over 100,000 charging piles and charging piles are uneven distributed that primarily concentrated in major cities and it is difficult to achieve the goal of 1 million charging piles in 2030. In addition, The German government have reduced subsidy for individuals purchasing electric vehicles at the end of 2023, which huge impact EV market negatively, the report from the Fortune mentioned that electric vehicle sales may decline by 14% in 2024. (RAUWALD, ECKL-DORNA, RAY, & Bloomberg, 2024)

This shows that German consumers are very sensitive to price. Many people are unwilling to pay the premium for electric vehicles. This situation has led many businesses have to provide discounts to consumers in order to sell the cars and that increases the bargaining power of

buyers indirectly. An autorotative Consumer study of 2023 by Deloitte reveals the main consumption propensity is the driving force of the market and the main concern for choosing the EV among driver include peer pressure, charging convenience, the Potential for extra taxes/levies applied to the internal combustion vehicles and the better driving experience and German consumers have sent a clear signal that there are still too many barriers to consumer charging - such as a lack of infrastructure, affordability and range. (Deloitte, 2023)

Thus, the level of bargaining power of buyer in German depends on the various factors. Firstly, the overall situation for EV purchase which mean if there is pressure to absolute should buy the EV car. However, from the Deloitte report we could see that the speed of shifts to EV in German seems moderate, 16% out of 1128 persons (sample size) would like to shift to EV as their next car which seems low, but this is the highest among other European countries in this survey. However, there are still 51% persons would still choose the internal combustion vehicles which also the second highest level among other testing countries. From the report we could the percentage of choose HEV and PHEV is 15% and 12% which are relate low than other countries in this survey, this means, when the German consumer would like to choose alternative than the internal combustion vehicles, more persons will consider the BEV than other type of cars. (Deloitte, 2023)

Combined with this implication with the current situation of German EV industry insufficient charging facilities, reduced subsidies and market slowdown as mentioned that prove that consumers will be more concerned about purchasing electric vehicles in this situation, so if the EU does not forbidden internal combustion vehicles directly or indirectly via tax in the future, the development of the entire market will continue to be slow result in the decline in sales which will actually increase barging power of buyer. However, the buyers' loyalty to the brand could also affect the their bargaining power thus the higher the brand awareness, the larger the number of loyal customers of a brand, the bargaining power of consumers will be relatively lower.

5.2.5 Competitive Rivalry

As discussed in the theoretical section, the threat from competitor usually discussing business competition, generally, the more competitor in the same market the more competition in this industry. To understand the opportunity and challenge in terms of competitive rivalry in the German EV market, it is worthy to investigate how many rivals, who they are, and what is their product compares with BYD offer.Competition in the German EV market intensifies since German electric vehicle subsidies reduce.

As outlined in the literature review, Tesla and Volkswagen emerged as the top-selling brands in the German EV market in 2023. Tesla model 3 and model Y are Tesla's two best-selling cars in 2023, with total sales starting from 61,483. Tesla Model 3 sales are 15,865 units, Model Y sales are 45,618 units, and Volkswagen ID 3 and ID 3 performance was also good, Volkswagen ID 3 sold 22,270 units and ID 4 sold 25,779 units. (Commision, 2024)

By Brand by model	Selling price in Germany (EURO)	WLTP (KM)	Acceleration from 0-100 km/h	Maximum speed (km/h)	Battery capicity (Kwh)	Power consumption
Tesla model Y performance	55990	533	3,7 seconds	250	82,1Kwh	25,5 kWh/100km³
Tesla Model 3	40990	547	6,1 seconds	228	57,5Kwh-75Kwh	13,5 kWh/100km³
BYD Tang	69000	400 combined ,528 in the city	4,6 seconds	180	80Kwh	23,8 kWh/100km³
BYD Seal standard	44990	550	5,9 seconds	180	84Kwh	16,6 kWh/100km³
VW ID3	39990	420	7 seconds	160	62kWh	13,9kWh/100km
VW ID4	43990	Up to 531	8,5 seconds	160	77Kwh	16,5kWh/100km
BYD Atto 3	39990	420(max 568 in the city)	7,3 seconds	160	60.5kWh	16kWh/100km
BYD Dolphin	32990	427	7 seconds	160	60.48 kWh	15.9kWh/100km

(Table 2, A comparison of Tesla, VW and BYD comparison, parameters come from BYD official website and Tesla official website)

If I only compare from the picture above, BYD TANG is 10,000 euros more expensive than Tesla model Y performance, acceleration from 0-100km/, Tesla model Y performance is 3.6

seconds which is faster than BYD TANG. Through the comparison with Tesla model 3 and BYD Seal as shown in the table, BYD does not seem to have a particularly big advantage either.Through the analysis from parameters above, the most competitive model of BYD is the BYD dolphin. BYD dolphin is newly launched BYD model with favorable price. BYD dolphin contains the most present advanced technology. therefore, when comparing BYD dolphin with VW ID3, the parameters such as WLTP, acceleration from 0-100 km/h, maximum speed and battery consumption as shown in table 2, they are almost the same. Moreover, in term of the body length of car the parameters are also similar, VW ID.3 is 4261mm and the wheelbase is 2765mm, while BYD Dolphin is 4150mm and the wheelbase is 2700mm. but BYD dolphin is almost 7000 euro cheaper than VW ID3 which could made BYD dolphin more attractive.

In order to find out how the competitive rivalry in the German EV market, I mainly compare several basic parameters between Germany's best-selling models of fully EV in 2023 with current BYD models sales models in Germany. Through this competition comparison, a low price could offer a distinct advantage for BYD while in the high-end price range do not possess any notable advantages based on these parameters. However, the parameters are considered here are only the basic parameters, many factors are not included, such as the appearance of the car, the core technology o, safety, and brand effect and so on which could have an influence on consumer's choice. If combined with Deloitte's report on barging power of buyer in the previous paragraph and the factors that price affect consumers' car purchases. The market for high-priced electric vehicles is saturated. Consumers prefer to have more high-performance electric vehicles in the low-price market to meet the needs of ordinary consumers, so this may be the reason why the German media is so optimistic about BYD Dolphin, and this model could be an opportunity for BYD is expanding its market share in Germany.

5.2.6 Partial conclusion

Throughout the elements of Porter's Five Force, this chapter helped BYD understand the structural dynamics of the German EV industry. In term of the threat of new entrants, the entry barriers is low due to its large initial investment and advanced technology requiring which means not every company has capability to entering this industry, especially for emerging brands which do not have the same resources and expertise to develop, the infrastructure to produce electric vehicles as established automobile manufacturers, establishing a foothold in the industry is difficult. The high enter barrier is an advantage for BYD because the more brands that come in, they will seize some market share.

Concerning the threat of substitutes, although electric vehicles are not the only solution in terms of environmental protection and emission reduction, there are also other alternative solutions, such as FCEV. However, throughout the analysis above, the development of FCEV is limited because there is no approve for FCEV are much better solution than BEV, at least for the next many years, the electric vehicle industry will still be the leader in the field of environmentally friendly and emission-reducing vehicles. Thus, the level of threat for the substitutes currently is moderate until the better alternative appears.

The level of bargaining power of suppliers could divide to the two group of suppliers, the core component supplier which means the product the supplier provide are significant for the EV company and will not be substituted easily, such as battery, in this type of suppliers their bargaining power is high. The other non-core component or could be substituted easily, the bargaining power will be lower. Since most of EV brands such as Tesla, VW, and Mercedes-Benz batteries are supplied by suppliers, such as Tesla's supplier CATL , in comparison, BYD has huge advantage in the supply chain because BYD has the vertical integrated supply chain which means most components of BYD's cars are manufactured in-house, as CEO of BYD Wang, Chuanfu mentioned in the interview that in addition to automobile tires and glass, BYD has independent R&D and production capabilities for almost all core components, which gives BYD a significant advantage in manufacturing costs.

The bargaining power of buyers in the German EV market is low for the individual purchase. This implication combined the current insufficient charging facilities , reduced subsidies in the German EV market which results in the decline in sales. If the sales continually bargaining power of buyer will increase. Beside that the buyers' loyalty to the brand could also affect their bargaining power and it means that the higher the brand awareness, the larger the number of loyal customers of a brand, the bargaining power of consumers will be relatively lower.

In terms of competitive rivalry, Competition in the German EV market intensifies since German electric vehicle subsidies reduce. throughout the competitor analysis conducted by comparing with the current sale model of BYD's in Germany with Tesla Y, Tesla 3, VWID3, VWID4. The analysis shows a low price model such as BYD Dolphin could offer a distinct advantage for BYD while in the high-end price range do not possess any notable advantages.

5.3 Organizational Resources

The RBV as Jay barney defined that focus on describing and identify company competitive advantage of company's internal resources and used as a managerial framework to development proper strategic. Resource-based View emphasizes that a company's competitive advantage comes from its unique and difficult-to-imitate resource combination. This paragraph will firstly identify the organizational resources relate the BYD's EV business, thereafter a VRIO framework will be applied to analyze these resources in order to assess whether these resources could be a competitive advantage of BYD based on value, rarity, inimitability and organization. (Barney, 1991)

To begin with, all resources relate to the BYD's EV business will be separate into distinctive resources and thresholds and by apply a 2x2 Matrix in table 3, the resource and competence within "threshold capability" of the matrix represent those resources of BYD are basic resources and competencies required to enter and compete in the industry. On the other

hand, those resources which are defined as “distinctive capabilities” means those resources and competences are unique for BYD in order to stand out in the German EV market. (Johnson, Scholes, & Regner, 2016)

	Resources	**Competences**
Threshold capabilities	• Raw materials • Suppliers • Buyers • Employees • Cash flow	• Knowledge of EV car manufacturing process • License • Ability to create and maintain customer relationship • Brand management
Distinctive capabilities	• Vertical integration supply chain • Strong Technology - Blade battery - Disu system - E platform - DMI（Dual Mode Intelligent System • Diversified product portfolio	• Highly capabilities technology innovation - Develop new car in 18 months • Cost control • Multi Brand Matrix - BYD - Denza - Yangwang - Fangchengbao

Table 3 2x2 Matrix of BYD internal resources and competences, 2x2 Matrix Model source from (Johnson, Scholes, & Whittington, 2008)

5.4 VRIO analysis of BYD's distinctive resources and competences

BYD has strong capabilities in independent research and development and has been committed to technological innovation in electric vehicles and has many leading patented technologies. For example, BYD electric vehicles use blade batteries and BYD launched the Disus system and E-platform which are distinguishes BYD among other EV brand.

5.4.1 Blade battery

BYD officially introduced the Blade Battery in March 2020 that utilizing lithium iron phosphate technology installed in the BYD HAN model initially. The Blade battery innovatively skips the module assembly process, significantly improving volume utilization. This design allows for fitting more battery cells within the same space, increasing energy density by over 50% compared to traditional battery packs.

BYD's Blade Battery has undergone rigorous safety tests, including the "Mount Everest" of battery safety tests – the nail penetration test (Apendix2), and extreme stress tests such as a 46-ton heavy vehicle rollover test. It boasts exceptional safety, strength, range, and lifespan. Safety features include seven-dimensional safety testing covering internal and external short circuits, overcharging, collisions, high voltage, connections, and hazardous gases.

In comparative tests conducted by BYD, the Blade Battery outperformed NCM Lithium and cubic lithium iron phosphate batteries in safety. After nail penetration, the Blade Battery remained unharmed, with no fire or smoke observed on the surface, and a surface temperature of 30-60°C. In contrast, NCM Lithium batteries experienced severe burning with surface temperatures exceeding 500°C, while cubic lithium iron phosphate batteries charred with observed smoke and surface temperatures ranging from 200-400°C. (BYD, BYD blade battery safer, better, stronger and more reliable, 2024)

5.4.2 E-platform

BYD's E-platform technology is a four-motor drive system capable of precisely controlling the dynamic of all four wheels, enabling safe driving in extreme conditions such as high-speed tire blowouts, emergency floating, and icy roads. Additionally, the technology enhances off-road capabilities. In terms of electronic control, the E-platform equips all models with a new generation SIC, boasting a high computational efficiency of up to 99.5% and control speed, increasing current output capacity by 50% for precise control of current output to all four driving wheels. On the perception level, the E-platform technology integrates various sensors, including traditional real-time data and intelligent driving sensor data such as cameras, LiDAR, millimeter-wave radar, and high-precision positioning. These sensors enable rapid and accurate reception of signals, supporting the identification, decision-making, and control of the four motors.

To achieve deep integration of perception, the E-platform technology utilizes a highly coordinated control architecture between the central computing platform and domain controllers. Through real-time exchange of perception information and control strategies via high-bandwidth, low-latency, and highly secure in-vehicle Ethernet, precise and diverse control of the four motors is achieved. As a result, the E-platform technology platform can independently adjust the dynamics of all four wheels at millisecond speed through a stronger perception power of the electric drive and electronic control system. This improves vehicle stability, reduces the risk of loss of control due to insufficient traction, and enhances handling stability, providing safety assurance for various usage scenarios. (Westerheide, 2024)

5.4.3 BYD DMI Technology

BYD DMI (Dual Mode Intelligent System) Technology is an efficient electronic control system serving as the dual-motor drive system for BYD vehicles. DMI Techonology offers pure electric mode with zero fuel consumption during city driving and high efficiency during high-speed driving. Overall, BYD DMI Technology provides low fuel consumption, high efficiency, and an exceptional driving experience. The Core Components including the EHS

Super Electric Hybrid System, Xiongyun Plug-in Hybrid Dedicated Efficient Engine, and DMI Super Hybrid-Specific Power Blade Battery in which BYD have Developed in-house. Key figures of BYD DMI Technology include electric-dominant hybrid technology that means based on a super electric hybrid system, shifting from traditional engine-reliant hybrid designs to prioritize electric power, significantly reducing fuel consumption. Dual Motor Drive that equipped with one motor each for the front and rear wheels, enhancing overall power output and fuel economy. Efficient Electronic Control System which automatically adjusts the output power of the front and rear wheel motors based on driving modes, optimizing power distribution for smoother driving and lower fuel consumption. With one motor for each wheel, DMI technology offers rapid response capabilities, allowing quick adjustments to different road conditions and emergencies, improving vehicle stability and safety, BYD DMI Technology has shown a strong safety performance. (Pandaily, 2024)

5.4.4 Disus system

In 2023, BYD officially unveiled the industry's first intelligent vehicle body control system exclusive to new energy vehicles, called the "Disus" system. This system, developed in-house by BYD, will be integrated into flagship models such as the Wangchao, Ocean, Yangwang,

and other personalized brand vehicles in the future. Disus represents a breakthrough in vehicle vertical control solutions, departing from fragmented approaches and foreign dependency. It achieves system-level integration, enabling comprehensive intelligent control of the vehicle body, including electrification and enhanced perception and decision-making capabilities. Furthermore, Disus enhances safety by minimizing vehicle posture changes, reducing the risk of rollovers, and protecting occupants during complex road conditions. Overall, Disus signifies a significant advancement in vehicle vertical control, elevating safety and stability standards for both occupants and vehicles. (BYD, BYD News, 2023)

When evaluating BYD’s strong technology innovation of their unique products such Blade battery, Disus system, E platform, DMI (Dual Mode Intelligent System) through the VRIO framework, firstly, all of them are very important and valuable resources for BYD, because these unique technologies enable BYD to become a leading company in EV industry. These resources is also the reason for BYD can create value for customers and attract customers. In term of rarity, they can all be considered rare resources, because the research and development of these products and systems involve many patented technologies pioneered by BYD.

When it comes to inimitability, some are difficult to imitate, such as blade battery, it is very challenging for the other contenders to acquire the same resource and capability such as BYD's blade battery which contains hundreds of patented technologies and so far no electric car manufacturer can manufacture and produce another blade battery since its launch in 2020. Thus, the blade battery is a rare and inimitable resource for BYD and the blade battery have been well organized because the bade battery has been used in almost every BYD's EV after it launched. Therefore, blade battery could be recognized as a sustained competitive advantage of BYD.

Another example is the Disus system launched by BYD in 2024. Disus system can considered as a sustained competitive advantage of BYD of its rarity and inimitability. Because the concept of Disus system has first been brainstormed in BYD's concept car in 2004, but it has been developed after 20 years of research and development with multiple patents which it is a technological breakthrough, therefore it will be difficult for other car brands to imitate within a certain period of time.

Regarding the rest of it as the DMI, E-Platform, they can be considered a rare resource for BYD due to not every electric vehicle on the market has the same product or technology. But they can only serve as temporary competitive advantage for BYD because they can be replaced, other Chinese brands such as Geely and Chery Automobile use also DMI gasoline-electric hybrid technology, and if the future market is transformed into pure electric, DMI will only be

a transitional product. In terms of E-platform is a system which developed by BYD in 2023, since other electric vehicle brands on the market also have their own electric vehicle software control systems and every company is constantly updating and developing in this field, therefore, E-platform may not be the unique and irreplaceable system.

5.4.5 Vertical integration supply chain

One of the key factors enabling BYD to control costs effectively is its vertical integration strategy, wherein the majority of the components used in BYD vehicles are produced in-house. As BYD CEO Wang Chuanfu introduced to the Chinese media that "BYD + Fuyao (glass) + Zhongce (rubber tires) + Baosteel (steel) = a complete automotive supply chain," which shows how powerful BYD's vertical supply chain is. Critical components such as batteries, motors, automotive electronics, molds, power semiconductors, and interior and exterior decorations, which constitute the highest-cost components in an electric vehicle. In 2020, BYD consolidated these operations into independent subsidiaries under the "Fudi" brand, including Fudi Power (motors, hybrid systems), Fudi Battery, Fudi Vision (lighting systems), Fudi Technology (automotive electronic components, chassis), and Fudi Mould. (Zhang, 2023)

BYD's vertical integration supply chain is one of the reasons why BYD can maintain its advantage in the EV industry and it could be seen as a sustained competitive advantage for BYD because of the following reasons. Firstly, BYD's vertical integration supply chain is an indispensable and valuable resource as it provides control over various stages of production, from raw materials to finished products, this control allows BYD to optimize efficiency, quality, and cost throughout the supply chain, contributing to its competitive advantage. Moreover, it may consider as relative rare resource, although the vertical integration is not uncommon, it is difficult to find other like BYD's extensive vertical integration across many different segments of its supply chain as battery production, vehicle manufacturing, and

component assembly. In term of imitability, replicating BYD's vertically integrated supply chain requires significant investment in infrastructure, technology, and organizational capabilities which may be challenging for other competitors to match the same level of vertical integration as BYD without similar resources and expertise. Finally, this resource is well organized because it constantly delivery the materials and components for manufacturing of BYD's EV and BYD has dedicated teams and processes in place to manage and optimize its vertically integrated supply chain, with a focus on coordination and synergy across different business units to continuously improve the quality and efficacity. (Cox, 2021)

5.4.6 Multi Brand Matrix and Diversified product portfolio

In BYD's annual report of 2023, it announced that BYD will adhere to the multi-brand matrix strategy in the electric vehicle business which means BYD will develop EV under all of BYD's brand include BYD, Denza, Yangwang, and Fangchengbao. These allow the company to cater to diverse customer segments and market preferences to bring business benefits diversified product portfolio, this strategy is an significant valuable for the BYD's development. However, many other EV company offer also diverse product portfolios and competitors may attempt to diversify their product portfolios to emulate BYD's approach, but achieving a similar level of diversification requires strategic foresight, investment, and organizational agility, for some companies may find this challenge than others. Therefore, Multi Brand Matrix and Diversified product portfolio may be competitive parity rather than sustained competitive advantage. (BYD, BYD periodic reports, 2023)

5.4.7 Strong Technology Innovation

BYD's strong R&D capabilities is one of the main reasons for BYD's ability to become the world's number one electric vehicle manufacturer. BYD's stipulation that it only takes 18

months to launch a new car, the rapid innovation enables BYD to stay competitive in the fast-paced industry and meet evolving consumers demands. It is a rare competence for BYD since it is very challenge for many companies to consistently develop new cars within such a short time. BYD's technology innovation may also be difficult to imitate due to its integration with the company's culture, processes, and resources. However, competitors may attempt to replicate this capability through investment in R&D and talent acquisition. Therefore, according to VRIO framework, this competence may be a temporary competitive advantage. (BYD, BYD periodic reports, 2023)

5.4.8 Cost Control:

Effective cost control is one of the important reasons for why BYD quickly seized the market. It enables BYD to maintain competitive pricing for its products, enhancing customer value and market competitiveness. BYD's ability to consistently manage costs efficiently may be rare within EV industry because it rely heavily on its strong ability of integrated vertical supply chain, as mentioned earlier it is very rare for company to obtain the same degree of integrated vertical supply chain, thus, it is difficult for company to achieve the same level of cost control.

Even though cost control is a common practice and cost control strategies are replicable but require a deep understanding of cost structures and efficient resource allocation, BYD's expertise in this area may be difficult for competitors to replicate without similar organizational capabilities plus cost control is one of most importance strategy in order to market development, thus, this competence could be considered a sustained competitive advantage for BYD. (Bambooworks, 2023)

5.4.9 Partial conclusion

The investigation of VRIO analysis of BYD’s resources and competences is summarized in table 4

Resources and competences	**V**	**R**	**I**	**O**	**Strategy implication**
Blade battery	Yes	Yes	Yes	Yes	Sustained competitve advantage
Disus system	Yes	Yes	Yes	Yes	Sustained competitve advantage
E-platform	Yes	Yes	No	Yes	Competitive parity
DMI (Dual Mode Intelligent System)	Yes	Yes	No	Yes	Competitive parity
Integrated vertical supply chain	Yes	Yes	Yes	Yes	Sustained competitve advantage
Diversed product portfolio	Yes	Yes	No	No	Competitive parity
Multi brand maxtrix	Yes	Yes	No	No	Competitive parity
Strong technology innovation	Yes	Yes	No	Yes	Competitive parity
Cost control	Yes	Yes	Yes	Yes	Sustained competitve advantage

Table 4: VRIO analysis of BYD's internal resources and competences

The table create a clearly understand that only if V(value), R(rarity), I(inimitability) and O(organized) are simultaneously satisfied four conditions can be considered as sustained competitive advantage. If only V (value) is met, but R (rarity) and/ or I (inimitability) are not met, these resources and capabilities can only be regarded as competitive parity. VRIO analysis helps BYD determine is strength in its resources and capabilities from the firm-level and could help BYD create the growth strategy and drive their vision in the future. (Enochson, 2022)

Chapter 6 Findings and Discussion

Throughout the analysis above, the multiple findings will be summarized and discussed in order to provide a comprehensive answer to the research question: *How can BYD adjust its business strategy adapting to the German EV market to increase sales?* This section delves into implications of BYD's business approach amidst the dynamic landscape of the German EV market and provide the insight into how BYD can leverage its resources to gain a competitive edge in order to increase the sales in Germany.

6.1 Institutional Opportunities and Challenges in the German EV Market

Germany's commitment to sustainability has implemented many favorable policies for the development of electric vehicles, as exemplified by initiatives such as the Climate Action Plan 2050 and the "Paris Agreement" presents a fertile ground for EV manufacturers like BYD. The nation's regulatory framework, including tax incentives and infrastructure development plans, provides a conductive environment for EV adoption.

However, challenges such as the shortage of charging infrastructure pose hurdles for market penetration, the German Ministry of Transport last year announced at the Munich Motor Show that it intends to support synthetic e-fuels as an option to reduce greenhouse gas emissions from passenger cars plus the electric vehicle market in 2024. There has been a slowdown, and some companies have delayed their pace of full electrification. For example, Mercedes-Benz media claimed at media this year that they will give up on full electrification and continue to develop fuel vehicles in the next 10 years. This is an opportunity for Mercedes-Benz to conduct current market research and technology maturity assessment. Mercedes-Benz's announcement seems to run counter to the EU's plan to ban internal-combustion cars in 2035. In other words, after Mercedes-Benz's analysis, it feels that the European Union plan to ban internal-combustion cars in 2035 will appear to be holding. (NIKKI, 2024)

If the EU once again delays the ban on internal combustion engine vehicles, it will have a great impact on Germany's ability to fully electrify its vehicles. After all, judging from Germany's current market share, internal combustion engines still account for half of the entire automotive market. BYD must navigate these complexities, eradicate policy changes and commission strategic goals to capitalize on the opportunities presented by Germany's EV landscape. (Padeanu, 2024)

6.2 BYD's Opportunities and Challenges in the German EV Industry

Throughout analysis of German EV industry based on Porter's Five Forces Model, the impression is that development prospects of the German EV industry are good in general for the following reasons. Firstly, given a strong awareness of environmental protection of Germans, the acceptance of EV increases gradually among the German population. Besides that, the EV industry has relatively high barriers to entry. There is little chance that other new energy sources seize the EV market in the short terms due to the huge initial investment and infrastructure requirements. For instance, there are only a few hundred FCEV are sold in the German market on the yearly base. (Erofeev, 2014)

Through the lens of Porter's Five Forces Model, it indicate that the huge advantages of BYD founded in the field of suppliers, as comparing with other EV brands, when most EV brands do not have technology to produce batteries by themselves, BYD has integrated vertical supply chain with enable BYD achieve more than 70% in-house production of their EV from raw materials, batteries and other important components to the final product. (BYD, BYD News, 2023)

The challenges also appear which maybe the reasons that affect BYD's sales. In terms of brand awareness and brand loyalty, as mentioned in the previous analysis, although BYD's awareness in Germany has been significantly improved as mentioned in the previous literature, and

German consumers acceptance of buying Chinese cars has increased, however, the German people are very loyal to German brands and if they want to change their car, most people will still buy the same brand as the old car. (Deloitte, 2023)Thus, it is importance for BYD focus on customer attraction and satisfaction in order to increase the customer base and customer loyalty. Leveraging its strengths and addressing these challenges is critical to BYD's success in Germany, BYD needs to adjust its strategy from the above aspects in order to increase the sales.

Moreover, another challenge is the current long delivery time, unlike Tesla and other big German brands, BYD currently does not have a factory in Germany and their factory in Hungary aims to be in production first at the end of 2023 in which means all the cars sold in the German market currently, are transported from China. Furtherly, German media reported

that due to the shortage of truck drivers in Germany, BYD's vehicles arriving at the port cannot be delivered to distributors, which makes the delivery time even longer. (Alim, Wright, & Campbell, 2024)

6.3 The strategic significance of BYD's resources and capabilities

The VRIO indicates which resources and capabilities of BYD are considered as sustained competitive advantages of BYD's EV ,and which ones can only be used as competitive parity VIRO analysis help BYD's adjust its business strategy better adapt to the German EV market in order to increase the sales. Blade batteries systems, Disus system, integrated vertical supply chain, and cost control are all BYD's key advantages for sustainable development. However, technology innovation, diverse product portfolio is also crucial of BYD's development.

Chapter 7 Recommendation

Based on the analysis and discussion provided, several recommendation are present in this section that can help BYD to adjust its business strategy customize to German EV market and increase its sales.

7.1 Strengthen the construction of charging infrastructure

By analyzing the current situation in the German EV market, solving the shortage of charging infrastructure is a top priority for BYD to speed up market penetration. BYD should work with government agencies and private stakeholders to speed up the installation of charging stations, especially in underserved areas. Investing in innovative charging solutions and providing incentives for the deployment of charging stations can enhance BYD's appeal to German consumers. For example, BYD's previous cooperation with Shell gave BYD customers a free charging card for charging at Shell charging stations. BYD can also introduce subsidies for the installation of some home charging piles to provide consumers with more convenient charging services.

7.2 Enhance brand awareness and loyalty

Building brand awareness and cultivating customer loyalty is critical to gaining market share in a competitive landscape. BYD should launch targeted marketing campaigns to showcase its technological innovation, product reliability and environmental credentials. Engaging with German consumers through social media, events and partnerships helps strengthen brand recognition and increase market share.For example, BYD becoming a partner of the European Cup 2024 is an opportunity to enhance brand awareness. In addition, BYD can also develop marketing strategies based on products. For example, BYD Seagull and Atto3 are high-performance and cost-effective daily family cars that the German public can afford. BYD can combine these characteristics of the models and match some representative local German celebrities or bloggers to cooperate with BYD, becoming BYD's publicity ambassador. Such a marketing strategy can help BYD enhance its brand image in market segments.

7.3 Adhere to diversified brands and product portfolios and focus on market segments

Consist on diversifying its product portfolio help BYD to meet the diverse consumer preferences in the German market. For instance, concerning the urban and eco-conscious families, offering a range of options, including electric SUVs, sedans and compact cars, can appeal to a wider audience and increase sales, while maintaining a focus on affordability and cost-effectiveness. Furthermore, because consumers are willing to buy high-performance and

affordable EV, BYD can capitalize on this consumer psychology focusing on the specific market segments can accelerate BYD's customer expansion. For example, BYD could put more budget investing on the products or segment that easiest to penetrate the market, such as low-price segments, increase its marketing and publicity efforts with celebrity endorsements, increase dealers' sales incentives and joint marketing activity in order to boost the sales.

Chapter 8 Conclusion

With the escalating global attention on climate change and environmental concerns, electric vehicles (EVs) emerge as a pivotal solution for reducing CO2 emissions. Consequently, there's been a noticeable surge in demand for electric vehicles globally. Germany, Europe's largest automobile market, which has made steadfast commitments to spearheading the production and adoption of EVs. The world leading EV manufacturer BYD has entered the German EV market within its passenger EV by the end of 2022 that aims to increase sales and capture 5-10% of the market. However, more than a year has passed, the poor sale performance of BYD's EV and the market share remains below 1%, trailing far behind like Tesla and Volkswagen. This prompts the question: *How can BYD adjust its business strategy adapting to the German EV market to increase sales?* Three sub-questions were constructed to investigate this research question. Through the analysis of first sub-question of what institutional opportunity and challenge does BYD face in Germany through the lens of IBV combining the investigation on

the existing literatures, it shows that there are policies, regulations, incentives as well as investment on EV infrastructures made by German government that evident Germany is deeply committed to EV development with clearly defined goals, plans. However, some incentive reduced in the end of 2023, it could have a negative impact on BYD's EV sales in Germany.

The second sub-question concerning the opportunities and challenges BYD face in the German EV market that is addressed through the application of Porter's Five Force. Opportunities are identified in terms of factors as threat of new entrants, threat of substitute and the barning power of supplier. Porter's Five Force revealed the entry barriers in German EV market are relatively high due to significant initial costs and technological requirements. The level of threat from substitute considered moderate in the German EV market as the EVs remains best solution for reducing CO2 emissions at least for the next many years until a better alternative emerges. Finally, in terms of bargaining power of supplier in German EV industry, BYD holds a significant advantage compared to its peers due to its vertically integrated supply chain, which allows for approximately 70% self-supply of their EV manufacturing, including batteries and motors.

Meanwhile, analysis also highlights challenges BYD encountered by BYD in the German EV market in terms of the buys bargaining power and competitive rivalry. Presently, consumers have limited recognition of the BYD brand, and BYD has not cultivate a loyal customer base yet. In addition, the competition in the German EV market is intense, comparing products with Tesla and Volkswagen. Through the comparison analysis of Tesla and Volkswagens best-selling models, it reveals that low- price segment is a distinct advantage for BYD while in the high-end price range do not possess any notable advantages. The last sub question "Which resources can BYD exploit to achieve a competitive advantage in the German EV market?" is addressed by applying the resources-based view framework, analyzing BYD's resources and competences in their EV business based on V (value), R (rarity), I (inimitability), and O

(organized). The result reveals the blade battery, disus system of BYD, integrated vertical supply chain and the strong cost control are found as most beneficial resources and competences in the organization's quest of gaining a competitive advantage.

Finally, based on the analysis and discussion, various recommendations are proposed to help BYD adjust its business strategy customize in the German EV market and increase sales. The recommendation focus on strengthen the construction of charging infrastructure, enhancement of brand awareness and loyalty, diversify the brands and product portfolios and develop different target on different market segments aim to focus on the core segment and product accelerate BYD's customer expansion and increase the sales. However, the German EV markis dynamic and competition is intense, in which presents both opportunities and challenges. BYD needs to continually adjust its strategies to keep pace with market changes to succeed.

Chapter 9 Further perspectives

There are some topics with this research that is worthy to further discussed, however due to time and page issues, these were not covered in this study.

Firstly, if the EV market continual to drop, how should BYD deal with it? As mentioned in this study, the German government and German companies now have time to launch full electrification, resulting in a 10-14% drop in electric vehicle sales this year. German companies such as Mercedes-Benz will continue to research internal combustion vehicles in the next 10 years. If this situation continues, it will have a huge negative impact on the BYD's EV development in Germany.

Secondly, the European Union has previously postponed full automation from 2030 to 2035. At the German Auto Show last year, several major German companies did not have new

products for pre-sale except concept cars. Does this mean that some German EV manufacturers have no new technological innovations in the field of electric vehicles currently? Coupled with the lack of infrastructure, will it delay the development of full electrification? Many people now believe that with the current development speed of electric vehicles, the EU is likely to delay the time for full electrification again? If this is really the case, how should BYD respond?

In addition, several other Chinese companies, including Zeeker from Geely and NIO, are also selling their EVs in the German market. Wang Chuanfu, the CEO of BYD expressed optimism about expanding overseas collectively in a previous interview. While they are competitors, as Chinese entities venturing abroad, how can BYD collaborate with them in overseas markets for mutual benefit and rapid enhancement of China's electric vehicle image among German consumers? This mutual support and cooperation are crucial considerations aligning with BYD's strategic objective of broadening its footprint in the German market.

Finally, this research solely examines segment of BYD's electronic passenger vehicles. However, it's worth noting that BYD's commercial vehicles, like BYD bus have made significant strides in penetrating the German as well as international markets. Additionally, BYD's sky rail initiative presents a promising, cost-effective urban transportation solution. How might the advancements in these sectors influence the development of BYD's electronic passenger vehicles in the German EV market in the future?

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